

When and Why SignSGD Outperforms SGD: A Theoretical Study Based on ℓ_1 -norm Lower Bounds

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Abstract

Sign-based optimization algorithms, such as SignSGD and Muon, have garnered significant attention for their remarkable performance in training large foundation models. Despite this empirical success, we still lack a theoretical understanding of when and why these sign-based methods outperform vanilla SGD. The core obstacle is that under standard smoothness and finite variance conditions, SGD is known to be minimax optimal for finding stationary points measured by ℓ_2 -norms, thereby fundamentally precluding any complexity gains for sign-based methods in standard settings. To overcome this barrier, we analyze sign-based optimizers leveraging ℓ_1 -norm stationarity, ℓ_∞ -smoothness, and a separable noise model, which can better capture the coordinate-wise nature of signed updates. Under this distinct problem geometry, we derive matched upper and lower bounds for SignSGD and explicitly characterize the problem class in which SignSGD provably dominates SGD. Specifically, we compare the *upper bound of SignSGD* with the *lower bound of SGD*, illustrating that SignSGD effectively reduces the complexity by a factor of d under *sparse noise*, where d is the problem dimension. Furthermore, we elevate this framework to the matrix domain, providing an equivalent optimal lower bound for the Muon optimizer, proving that extending the sign operator to matrices preserves this optimal scaling with dimensionality. Finally, we bridge our theoretical bounds to practice, demonstrating that the theoretical superiority of SignSGD accurately predicts its faster convergence during the pretraining of a 124M parameter GPT-2 model. Code is available at https://github.com/Dingzhen230/SignSGD_Outperforms_SGD.

1 Introduction

Efficient and scalable stochastic optimization algorithms play an indispensable role in the huge success of large foundation models [Devlin et al., 2019, Brown et al., 2020, Achiam et al., 2023, Touvron et al., 2023, Team et al., 2023]. Among these, sign-based optimization algorithms, such as SignSGD [Bernstein et al., 2018] and Muon [Jordan et al., 2024], have attracted increasing focus due to substantial empirical edges over stochastic gradient descent (SGD) [Robbins and Monro, 1951]. SignSGD leverages 1-bit signed gradient to update the model, which naturally enjoys many favorable empirical properties in distributed environments [Bernstein et al., 2019] and low-precision training regimes [Yu et al., 2026a]. Muon optimizes using the matrix sign operator, implemented via Newton–Schulz iterations [Kovarik, 1970, Björck and Bowie, 1971]. Numerous empirical studies [Liu et al., 2025, Shah et al., 2025, Wen et al., 2025, Semenov et al., 2025] have shown its consistent speedup over AdamW [Loshchilov and Hutter, 2019], and it has become a new industrial paradigm for pretraining massive-scale large language models (LLMs) [Team et al., 2025, 2026, Zeng et al., 2025, 2026, Cheng et al., 2026, DeepSeek-AI, 2026].

何时以及为何 SignSGD 超越 SGD：基于 ℓ_1 -范数下界的理论研究

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摘要

基于符号的优化算法，如 SignSGD 和 Muon，因其在大规模基础模型训练中的卓越性能而备受关注。尽管取得了这种经验上的成功，但我们仍然缺乏对何时以及为何这些基于符号的方法超越普通 SGD 的理论理解。核心障碍在于，在标准的平滑性和有限方差条件下，已知 SGD 在以 ℓ_2 -范数衡量驻点查找方面是 minimax 最优的，从而从根本上排除了基于符号方法在任何标准设置中的复杂度提升。为了克服这一障碍，我们分析了利用 ℓ_1 -范数驻点性、 ℓ_∞ -平滑性和可分离噪声模型的基于符号的优化器，这些方法能更好地捕捉符号更新的坐标性质。在这个独特的问题几何形状下，我们推导出 SignSGD 匹配的上限和下限，并明确刻画了 SignSGD 可证明超越 SGD 的问题类别。具体而言，我们比较了 SignSGD 的上限与 SGD 的下限，表明 SignSGD 在稀疏噪声条件下有效降低了复杂度，其中 d 是问题维度。此外，我们将这一框架扩展到矩阵域，为 Muon 优化器提供了一个等效的最优下界，证明将符号算子扩展到矩阵上可以保持这种与维度的最优缩放关系。最后，我们将理论界限与实践联系起来，证明 SignSGD 的理论优越性准确预测了其在 124M 参数 GPT-2 模型预训练过程中的更快收敛。代码可在 https://github.com/Dingzhen230/SignSGD_超越_SGD。

1 引言

高效且可扩展的随机优化算法在大规模基础模型 [Devlin 等人, 2019, Brown 等人, 2020, Achiam 等人, 2023, Touvron 等人, 2023, Team 等人, 2023] 的巨大成功中发挥着不可或缺的作用。在这些算法中，基于符号的优化算法，如 SignSGD [Bernstein 等人, 2018] 和 Muon [Jordan 等人, 2024]，由于在随机梯度下降 (SGD) [Robbins 和 Monro, 1951] 之上具有显著的实证优势而受到越来越多的关注。SignSGD 利用 1 位符号梯度来更新模型，在分布式环境中自然具有许多有利的实证特性 [Bernstein 等人, 2019] 以及低精度训练机制 [Yu 等人, 2026a]。Muon 使用矩阵符号算子进行优化，通过牛顿-舒尔茨迭代 [Kovarik, 1970, Björck 和 Bowie, 1971] 实现。大量实证研究 [Liu 等人, 2025, Shah 等人, 2025, Wen 等人, 2025, Semenov 等人, 2025] 表明其始终如一地比 AdamW [Loshchilov 和 Hutter, 2019] 加速，并已成为大规模大语言模型 (LLMs) 预训练的新工业范式 [Team 等人, 2025, 2026, Zeng 等人, 2025, 2026, Cheng 等人, 2026, DeepSeek-AI, 2026]。

The empirical success of sign-based optimizers has motivated a fruitful line of work that attempts to justify their advantages theoretically. From an optimization theory perspective, the convergence of sign-based methods under various assumptions has been extensively studied [Bernstein et al., 2018, 2019, Safaryan and Richtárik, 2021, Sun et al., 2023, Jiang et al., 2025b, Yu et al., 2026b]. However, a complete theoretical understanding of why sign-based optimizers outperform SGD remains elusive. The biggest obstacle preventing any theoretical advances is that *SGD is already worst-case optimal when the objective function is smooth and stochastic gradients are unbiased with bounded variance* [Arjevani et al., 2023]. Under these standard assumptions, any first-order algorithm requires at least $\Omega(\epsilon^{-4})$ queries to find an ϵ -stationary point measured by the ℓ_2 -norm, a complexity tightly matched by SGD [Ghadimi and Lan, 2013]. Consequently, evaluating the performance of sign-based methods under the aforementioned problem class and ℓ_2 -stationary measure inherently fails to deliver any provable gains we desire.

In this paper, we systematically investigate when and why sign-based methods can outperform SGD by delving into a new problem class and performance measure that depart from the canonical setting in Arjevani et al. [2023]. Specifically, instead of the traditional ℓ_2 -geometry based on ℓ_2 -smoothness, finite variance, and ℓ_2 -stationarity, we consider an ℓ_∞ -geometry with ℓ_∞ -smoothness, coordinate-wise finite variance, and ℓ_1 -stationarity. This geometry shift is motivated by the insight that sign-based updates are intrinsically aligned with ℓ_∞ -geometry [Balles et al., 2020, Bernstein and Newhouse, 2024, Xie et al., 2025a,b]. Under this alternative geometry, we provide the first rigorous characterization of a problem class in which SignSGD provably outperforms SGD.

To clarify this geometric shift, it is essential to highlight the distinction between these conditions. Standard ℓ_2 -smoothness assumes the loss landscape curves uniformly across all directions, while ℓ_∞ -smoothness bounds the gradient change based on the maximum coordinate-wise distance, providing a hypercube-based geometric assumption that aligns with coordinate-wise, scaled updates of sign-based methods. Define $D_f(\mathbf{x}, \mathbf{y}) := f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle)$ as Bregman Divergence of f , the above two smooth model can be expressed as

$$|D_f(\mathbf{x}, \mathbf{y})| \leq \frac{L_2}{2} \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \ell_2\text{-smoothness}; \quad |D_f(\mathbf{x}, \mathbf{y})| \leq \frac{L_\infty}{2} \|\mathbf{y} - \mathbf{x}\|_\infty^2, \quad \ell_\infty\text{-smoothness}.$$

Furthermore, we adopt a separable noise model where variance is tracked independently for each coordinate via a vector $\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \dots, \sigma_d]$. This fine-grained characterization is more suitable for analyzing the highly imbalanced noise prevalent in modern deep learning [Sagun et al., 2017, Zhu et al., 2019, Zhang et al., 2020, Pan et al., 2022, Wu et al., 2022, Pan et al., 2024]:

$$\begin{aligned} \mathbb{E} [\|\mathbf{g}_t^b - \nabla f(\mathbf{x}_t)\|_2^2 | \mathcal{F}_{t-1}] &\leq \sigma^2, \quad \text{Standard noise model;} \\ \forall i \in [d], \mathbb{E} [\left| \mathbf{g}_{t,i}^b - \nabla_i f(\mathbf{x}_t) \right|^2 | \mathcal{F}_{t-1}] &\leq \sigma_i^2, \quad \text{Separable noise model.} \end{aligned}$$

Under this refined geometry, we develop a clean, self-contained lower bound analysis for SignSGD. We prove an ℓ_1 -norm lower bound that matches its upper bound, thereby giving a tight characterization of its convergence rate. We further derive an ℓ_1 -norm lower bound for SGD under the same setting. Comparing the *upper bound of SignSGD* with the *lower bound of SGD* reveals a strict separation: when the coordinate-wise noise is sparse or highly heterogeneous, SignSGD achieves a better dimension dependence. Finally, we extend this framework to matrix optimization and establish the first lower bound for Muon measured by the nuclear norm with matching upper bounds.

The main contributions of this paper are summarized as follows.

- **Tight Bounds for SignSGD:** Let L_∞ and $\boldsymbol{\sigma}$ denote the ℓ_∞ -smooth Lipschitz constant and the noise variance vector, respectively. We establish that the convergence rate of SignSGD with a constant step size is: $\mathbb{E} [\min_t \|\nabla f(\mathbf{x}_t)\|_1] = \Theta \left(\sqrt{L_\infty \Delta / N} + (\|\boldsymbol{\sigma}\|_1^2 L_\infty \Delta / N)^{\frac{1}{4}} \right)$.
- **Provable Dimensional Gains of SignSGD over SGD:** By comparing SignSGD’s *upper bound* with SGD’s *lower bound*, we prove that SignSGD achieves a strictly superior dimension dependence. We show that when the noise $\boldsymbol{\sigma}$ is sparse, SignSGD’s complexity can be d times better than that of SGD. We validate these theoretical findings by demonstrating accelerated convergence during the pretraining of a 124M-parameter GPT-2 model.
- **Unified Extension to Matrix Optimizers:** We elevate our theoretical framework into the matrix domain, mapping separable operations to orthogonalized matrix steps. By applying

基于符号的优化器的实证成功激发了一系列富有成效的研究工作，试图从理论上证明它们的优势。从优化理论的角度来看，在各种假设下基于符号方法的收敛性已被广泛研究 [Bernstein 等人, 2018, 2019, Safaryan 和 Richtárik, 2021, Sun 等人, 2023, Jiang 等人, 2025b, Yu 等人, 2026b]。然而，为什么基于符号的优化器优于 SGD 的完整理论理解仍然难以捉摸。阻碍任何理论进展的最大障碍是，当目标函数光滑且随机梯度无偏且方差有界时，SGD 已经是最坏情况最优的 [Arjevani 等人, 2023]。在这些标准假设下，任何一阶算法至少需要 $\Omega(\epsilon^{-4})$ 次查询才能找到由 ℓ_2 -范数测量的 ϵ -稳定点，其复杂度与 SGD [Ghadimi 和 Lan, 2013] 紧密匹配。因此，在上述问题类别和 ℓ_2 -稳定度测量下评估基于符号方法的表现本质上无法提供我们期望的可证明收益。

在本文中，我们系统地研究了何时以及为何符号方法能够优于SGD，通过深入研究一个新的问题类别和性能度量，这些都与 Arjevani等人 [2023]中的规范设置不同。具体来说，我们不再考虑基于 ℓ_2 -几何的 ℓ_2 -光滑度、有限方差和 ℓ_2 -平稳性，而是考虑一个 ℓ_∞ -几何，具有 ℓ_∞ -光滑度、坐标有限方差和 ℓ_1 -平稳性。这种几何转变的动机在于，符号更新与 ℓ_∞ -几何 [Balles等人, 2020, Bernstein和Newhouse, 2024, Xie等人, 2025a,b]中的内在一致。在这种替代几何下，我们首次对一个问题类别进行了严格的刻画，其中SignSGD可证明优于SGD。

为了阐明这种几何转变，必须强调这些条件之间的区别。标准的 ℓ_2 -光滑度假设损失曲面在所有方向上均匀弯曲，而 ℓ_∞ -光滑度基于最大坐标距离来限制梯度变化，提供了一种基于超立方体的几何假设，这与符号方法的坐标按比例更新的方式相一致。定义 $D_f(\mathbf{x}, \mathbf{y}) := f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle)$ 为 f 的Bregman散度，上述两种光滑模型可以表示为

$$|D_f(\mathbf{x}, \mathbf{y})| \leq \frac{L_2}{2} \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \ell_2\text{-smoothness}; \quad |D_f(\mathbf{x}, \mathbf{y})| \leq \frac{L_\infty}{2} \|\mathbf{y} - \mathbf{x}\|_\infty^2, \quad \ell_\infty\text{-smoothness}.$$

此外，我们采用可分离噪声模型，其中通过向量独立跟踪每个坐标的方差 $\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \dots, \sigma_d]$ 。这种细粒度表征更适合分析现代深度学习中普遍存在的高度不平衡噪声 [Sagun 等人, 2017, Zhu 等人, 2019, Zhang 等人, 2020, Pan 等人, 2022, Wu 等人, 2022, Pan 等人, 2024]:

$$\begin{aligned} \mathbb{E} [\|\mathbf{g}_t^b - \nabla f(\mathbf{x}_t)\|_2^2 | \mathcal{F}_{t-1}] &\leq \sigma^2, \quad \text{Standard noise model;} \\ \forall i \in [d], \mathbb{E} [\left| \mathbf{g}_{t,i}^b - \nabla_i f(\mathbf{x}_t) \right|^2 | \mathcal{F}_{t-1}] &\leq \sigma_i^2, \quad \text{Separable noise model.} \end{aligned}$$

在此精细几何下，我们为 SignSGD 开发了一个干净、自包含的下界分析。我们证明了一个 ℓ_1 -范数下界，它与上界相匹配，从而给出了其收敛率的精确表征。我们进一步在相同设置下为 SGD 推导了一个 ℓ_1 -范数下界。比较 SignSGD 的上界与 *SGD* 的下界揭示了一种严格分离：当坐标噪声稀疏或高度异构时，SignSGD 实现了更好的维度依赖性。最后，我们将此框架扩展到矩阵优化，并建立了第一个以核范数为标准的 Muon 下界，其上界与之匹配。

本文的主要贡献总结如下。

- **SignSGD的紧致界：** 令 L_∞ 和 $\boldsymbol{\sigma}$ 分别表示 ℓ_∞ -光滑Lipschitz常数和噪声方差向量。我们证明了具有固定步长的SignSGD的收敛速度为： $\mathbb{E} [\min_t \|\nabla f(\mathbf{x}_t)\|_1] = \Theta \left(\sqrt{L_\infty \Delta / N} + (\|\boldsymbol{\sigma}\|_1^2 L_\infty \Delta / N)^{\frac{1}{4}} \right)$ 。
- **SignSGD相较于SGD的可证明维度增益：** 通过比较SignSGD的上界与SGD的下界，我们证明了SignSGD实现了更严格的维度依赖性。我们表明，当噪声 $\boldsymbol{\sigma}$ 是稀疏时，SignSGD的复杂度可以 d 倍优于SGD。我们通过展示在124M参数GPT-2模型的预训练过程中加速收敛来验证这些理论发现。
- **统一扩展至矩阵优化器：** 我们将理论框架提升至矩阵领域，将可分离操作映射为正交化的矩阵步骤。通过应用

spectral norm smoothness and matrix variance, we provide equivalent upper and lower bounds for the recently proposed Muon optimizer [Jordan et al., 2024], proving for the first time that Muon handles large-scale matrix parameters just as effectively as SignSGD handles flat vectors.

2 Related Work

Sign-based methods: vector optimizers The idea of using only gradient signs dates back at least to RProp [Riedmiller and Braun, 1993], while the modern stochastic formulation was introduced by Bernstein et al. [2018], who studied SignSGD and its momentum variant Signum for smooth non-convex optimization. Since SignSGD replaces the stochastic gradient by its coordinate-wise sign, its natural stationarity measure is ℓ_1 -norm rather than the usual ℓ_2 -norm. This geometry is also closely tied to communication efficiency, as one-bit sign updates and majority vote make SignSGD attractive in distributed learning environments [Bernstein et al., 2019, Jin et al., 2020, Safaryan and Richtárik, 2021].

A fundamental limitation of vanilla sign updates appears in non-smooth optimization. Even for simple convex non-smooth objectives, SignSGD can fail to converge because the discontinuity of the sign map may repeatedly select an unfavorable subgradient direction [Karimireddy et al., 2019, Xiao et al., 2023]. Error feedback repairs this issue by accumulating and correcting compression errors [Karimireddy et al., 2019]. More recently, StoSignSGD [Yu et al., 2026a] takes a different route by injecting unbiased structural stochasticity directly into the sign conversion. Its coordinate-wise max-buffer controls the stochastic sign scale, and its noise level is coupled with the gradient signal. This design preserves the numerical simplicity of sign updates while resolving the non-smooth non-convergence pathology of deterministic SignSGD.

In smooth stochastic optimization, a separate line of work sharpens the convergence theory of sign-based methods through momentum-based analyses [Sun et al., 2023, Jiang et al., 2025b], stochastic sign mechanisms [Jin et al., 2020, Safaryan and Richtárik, 2021], variance reduction [Chzhen and Schechtman, 2023, Jiang et al., 2024], and heavy-tailed noise analyses [Kornilov et al., 2025, Yu et al., 2026b]. Sign-based vector optimizers are also closely related to adaptive methods. Several works interpret the optimization dynamics of Adam [Kingma and Ba, 2015] through signed or normalized update directions [Balles and Hennig, 2018, Crawshaw et al., 2022, Kunstner et al., 2023, Peng et al., 2025], while Lion [Chen et al., 2023] has emerged as a representative momentum-based sign optimizer with strong empirical performance and growing theoretical support [Chen et al., 2024, Dong et al., 2024, Jiang and Zhang, 2025, Sfyraki and Wang, 2025, Yu et al., 2026b]. Most relevant to our motivation, Yu et al. [2026b] demonstrate that SignSGD- and Lion-type methods can provably outperform AdamW, especially in LLM training regimes where the noise is heavy-tailed [Zhang et al., 2020].

Sign-based methods: matrix optimizers Matrix sign optimizers extend the sign-based philosophy from coordinate-wise vector updates to structured matrix updates. The representative example is Muon [Jordan et al., 2024], which applies an orthogonalized matrix sign direction: for a matrix gradient $G = U\Sigma V^\top$, the update direction is $\text{msign}(G) = UV^\top$. In practice, this matrix sign operation is efficiently approximated by Newton–Schulz iterations [Kovarik, 1970, Björck and Bowie, 1971]. Subsequent variants, such as MuonLight, incorporate Nesterov momentum, learning-rate alignment, and implementation refinements, and have been used or benchmarked in massive-scale language model training [Liu et al., 2025, Zeng et al., 2025, Team et al., 2025, Wen et al., 2025, Semenov et al., 2025, DeepSeek-AI, 2026]. These works suggest that matrix-level normalization can provide optimization benefits beyond coordinate-wise sign updates.

Theoretical understanding of Muon and related matrix optimizers is still developing. Existing studies analyze Muon from several complementary perspectives, including spectral and nuclear-norm geometry, normalized matrix descent, preconditioning, connections to Adam-type methods, and empirical justifications for its superiority [Li and Hong, 2025, Shen et al., 2025, Chang et al., 2025, Si et al., 2025, Sfyraki and Wang, 2025, Chen et al., 2025, Huang et al., 2025, Li et al., 2025b, Qian et al., 2025, Tveit et al., 2025, Page et al., 2025, Mehta et al., 2025, Vasudeva et al., 2025a,b, Frans et al., 2025, Pan et al., 2025, Wang et al., 2025, Zhang and Gao, 2025, Zhang et al., 2025, Su, 2025, Crawshaw et al., 2025, Ma et al., 2026, Du and Su, 2026].

Lower complexity bounds Complexity lower bounds are the standard tool for determining whether an apparent algorithmic improvement is genuine or merely an artifact of analysis. The oracle-

谱范数平滑性和矩阵方差，我们为最近提出的 Muon 优化器提供了等价的上界和下界 [Jordan 等人, 2024], 首次证明了 Muon 处理大规模矩阵参数的效果与 SignSGD 处理扁平向量的效果一样出色。

2 相关工作

符号方法：向量优化器 仅使用梯度符号的想法至少可以追溯到 RProp [Riedmiller 和 Braun, 1993], 而现代随机公式由 Bernstein 等人 [2018], 引入，他们研究了 SignSGD 及其动量变体 Signum 用于平滑非凸优化。由于 SignSGD 用其坐标符号替换了随机梯度，其自然平稳性度量是 ℓ_1 -范数而不是通常的 ℓ_2 -范数。这种几何形状也与通信效率密切相关，因为一位符号更新和多数投票使 SignSGD 在分布式学习环境中具有吸引力 [Bernstein 等人, 2019, Jin 等人, 2020, Safaryan 和 Richtárik, 2021]。

纯符号更新的一个基本局限性体现在非光滑优化中。即使对于简单的凸非光滑目标，SignSGD 也可能无法收敛，因为符号映射的间断性可能会反复选择不利的子梯度方向 [Karimireddy 等人, 2019, Xiao 等人, 2023]。误差反馈通过累积和纠正压缩误差来修复这个问题 [Karimireddy 等人, 2019]。最近，StoSignSGD [Yu 等人, 2026a] 采取了不同的方法，通过直接将无偏的结构化随机性注入符号转换中。其坐标方向的 max 缓冲器控制随机符号尺度，其噪声水平与梯度信号耦合。这种设计保留了符号更新的数值简单性，同时解决了确定性 SignSGD 的非光滑非收敛病理。

在平滑随机优化中，一条独立的研究路线通过基于动量的分析，提升了基于符号方法的收敛理论 [Sun 等人, 2023, Jiang 等人, 2025b]。随机符号机制 [Jin 等人, 2020, Safaryan 和 Richtárik, 2021], 方差降低 [Chzhen 和 Schechtman, 2023, Jiang 等人, 2024], 以及重尾噪声分析 [Kornilov 等人, 2025, Yu 等人, 2026b]。基于符号的向量优化器也与自适应方法密切相关。几项工作通过符号或归一化更新方向解释了 Adam [Kingma 和 Ba, 2015] 的优化动态 [Balles 和 Hennig, 2018, Crawshaw 等人, 2022, Kunstner 等人, 2023, Peng 等人, 2025], 而 Lion [Chen 等人, 2023] 已成为一种具有强大经验性能和日益增长的理论支持的基于动量的符号优化器 [Chen 等人, 2024, Dong 等人, 2024, Jiang 和 Zhang, 2025, Sfyraki 和 Wang, 2025, Yu 等人, 2026b]。与我们动机最相关的是，Yu 等人. [2026b] 证明了 SignSGD 和 Lion 类方法可以证明优于 AdamW，特别是在噪声为重尾的 LLM 训练机制中 [Zhang 等人, 2020]。

符号方法：矩阵优化器 矩阵符号优化器将符号方法的哲学从坐标向量更新扩展到结构化矩阵更新。代表性例子是 Muon [Jordan 等人 2024], 它应用正交化的矩阵符号方向：对于矩阵梯度 $G = U\Sigma V^\top$ ，更新方向是 $\text{msign}(G) = UV^\top$ 。在实践中，这个矩阵符号操作通过牛顿 - 舒尔茨迭代 [Kovarik, 1970, Björck 和 Bowie, 1971] 高效近似。后续变体，如 MuonLight，结合了 Nesterov 动量、学习率对齐和实现优化，已被用于或在大规模语言模型训练中基准测试 [Liu 等人, 2025, Zeng 等人, 2025, Team 等人, 2025, Wen 等人, 2025, Semenov 等人, DeepSeek-AI, 2026]。这些工作表明，矩阵级归一化可以提供比坐标符号更新更优的优化收益。

对 Muon 和相关矩阵优化器的理论理解仍在发展中。现有研究从多个互补的视角分析了 Muon，包括谱范数和核范数几何、归一化矩阵下降、预处理、与 Adam 类方法的关系，以及对其优越性的经验论证 [Li 和 Hong, 2025, Shen 等人, 2025, Chang 等人, 2025, Si 等人, 2025, Sfyraki 和 Wang, 2025, Chen 等人, 2025, Huang 等人, 2025, Li 等人, 2025b, Qian 等人, 2025, Tveit 等人, 2025, Page 等人, 2025, Mehta 等人, 2025, Vasudeva 等人, 2025a,b, Frans 等人, 2025, Pan 等人, 2025, Wang 等人, 2025, Zhang 和 Gao, 2025, Zhang 等人, 2025, Su, 2025, Crawshaw 等人, 2025, Ma 等人, 2026, Du 和 Su, 2026]。

下界复杂度 复杂度下界是确定一个明显的算法改进是真实的还是仅仅是分析结果的一种标准工具。神谕-

complexity viewpoint goes back to the classical framework of [Nemirovski and Yudin \[1983\]](#), and was later developed extensively for stochastic convex optimization [[Agarwal et al., 2012](#)]. These results identify how noise, dimension, and geometry constrain the best possible rates in stochastic first-order optimization.

For smooth non-convex optimization, the classical target is an ϵ -stationary point measured by $\|\nabla f(x)\|_2$. In the deterministic setting, [Carmon et al. \[2020, 2021\]](#) established sharp lower bounds for finding stationary points of smooth high-dimensional functions, while [Chewi et al. \[2023\]](#) showed that even in one dimension the query complexity depends delicately on whether the algorithm is deterministic or randomized and whether it can access first-order or zeroth-plus-first-order information. In the stochastic setting, [Ghadimi and Lan \[2013\]](#) gave the standard $O(\epsilon^{-4})$ upper bound for SGD under unbiased bounded-variance gradients, and subsequent lower-bound work showed that this rate is essentially unavoidable. In particular, [Drori and Shamir \[2020\]](#) studied worst-case lower bounds for SGD itself, and [Arjevani et al. \[2023\]](#) proved that any stochastic first-order method requires $\Omega(\epsilon^{-4})$ stochastic-gradient queries under the standard bounded-variance model, with a corresponding $\Omega(\epsilon^{-3})$ lower bound under mean-squared smoothness. Thus, in the usual ℓ_2 -geometry, SGD is minimax optimal, and one should not expect a worst-case improvement without changing the geometry, the oracle model, or the stationarity criterion.

A more recent line of work asks whether adaptive or sign-based methods can provably improve over SGD once the problem structure is refined. [Jiang et al. \[2025a\]](#) show that this is possible for AdaGrad under coordinate-wise smoothness and coordinate-wise noise assumptions: by adopting an ℓ_1 -stationarity measure, they prove both an AdaGrad upper bound and an SGD lower bound, yielding regimes where AdaGrad improves over SGD by a factor of the dimension. Complementarily, [Crawshaw and Liu \[2025\]](#) study lower bounds for adaptive gradient algorithms under (L_0, L_1) -smoothness, showing that several AdaGrad variants necessarily incur higher-order dependence on the relaxed-smoothness parameters.

3 SignSGD: Upper Bound and Lower Bound

In this section, we give the first tight characterization of the convergence of SignSGD.

3.1 Notations and Assumptions

We write $[T]$ for $\{1, 2, \dots, T\}$, $|\mathbf{x}|$ for element-wise absolute value of $\mathbf{x} \in \mathbb{R}^d$. The $\mathbf{L}$ -weighted vector norm for $\mathbf{L} \in \mathbb{R}_+^d$ is defined as $\|\mathbf{x}\|_{\mathbf{L}}^2 := \mathbf{x}^\top \text{diag}(\mathbf{L})\mathbf{x}$. For vector descent methods, we study the optimization problem $\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$, where $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable. Given a point $\mathbf{x} \in \mathbb{R}^d$, we can only access the gradient $\nabla f(\mathbf{x}) = [\nabla_1 f(\mathbf{x}), \dots, \nabla_d f(\mathbf{x})] \in \mathbb{R}^d$ in a noisy manner which we will later define in Assumption 2a. We use $\text{sign}(\cdot)$ to denote the sign operator, and WLOG, use the convention $\text{sign}(0) = 0$, which our lower bound analysis may encounter. Below, we list some necessary assumptions.

Assumption 1a (Lower bounded objective). The function f is bounded from below. There exists $f^* > -\infty$ such that $f(\mathbf{x}) \geq f^*$, for all $\mathbf{x} \in \mathbb{R}^d$. We further denote $\Delta = f(\mathbf{x}_0) - \inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$.

Assumption 2a (Separable noise model). At step t we observe a mini-batch of mutually independent gradients $g_t = \{\mathbf{g}_t^1, \dots, \mathbf{g}_t^B\}$ satisfying $\mathbb{E}[\mathbf{g}_t^b | \mathcal{F}_{t-1}] = \nabla f(\mathbf{x}_t), \forall b \in [B]$ where $\mathcal{F}_t = \sigma(g_1, \dots, g_t)$ denotes the natural filtration. Moreover, the coordinate-wise conditional variance is bounded separably: there exist non-negative constants $\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \dots, \sigma_d]$ such that

$$\mathbb{E} \left[\left| \mathbf{g}_{t,i}^b - \nabla_i f(\mathbf{x}_t) \right|^2 \middle| \mathcal{F}_{t-1} \right] \leq \sigma_i^2, \quad \forall i \in [d], \forall b \in [B].$$

Assumption 1a is standard and necessary for stochastic non-convex optimization [[Arjevani et al., 2023](#)]. Assumption 2a is widely used in analysis of adaptive methods like AdaGrad [[Duchi et al., 2011](#)], and sign-based methods like SignSGD [[Bernstein et al., 2018](#), [Li et al., 2025a](#), [Jiang et al., 2025a](#), [Yu et al., 2026b](#)].

Assumption 3a (ℓ_∞ -smoothness). The objective function f is ℓ_∞ -smooth if there exists non-negative constant L_∞ such that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we have

$$|f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle)| \leq \frac{L_\infty}{2} \|\mathbf{y} - \mathbf{x}\|_\infty^2.$$

复杂度视角追溯到经典框架的`Nemirovski and Yudin`, 后来在随机凸优化方面被`Agarwal et al.`等人广泛发展。这些结果确定了噪声、维度和几何形状如何限制随机一阶优化中的最佳可能速率。

对于平滑非凸优化, 经典目标是一个 ϵ -驻点, 该驻点由 $\|\nabla f(x)\|_2$ 衡量。在确定性场景下, [Carmon 等人 \[2020, 2021\]](#) 为寻找高维平滑函数的驻点建立了精确的下界, 而[Chewi 等人 \[2023\]](#) 表明, 即使在单维情况下, 查询复杂度也微妙地取决于算法是确定性还是随机性, 以及它能否获取一阶或零阶加一阶信息。在随机性场景下, [Ghadimi 和 Lan \[2013\]](#)给出了 SGD 在无偏有界方差梯度下的标准 $O(\epsilon^{-4})$ 上界, 后续的下界工作表明该速率本质上不可避免。特别是, [Drori 和 Shamir \[2020\]](#) 研究了 SGD 本身的最坏情况下下界, 而[Arjevani 等人 \[2023\]](#) 证明任何随机一阶方法在标准有界方差模型下都需要 $\Omega(\epsilon^{-4})$ 随机梯度查询, 对应地, 在均方平滑性下有相应的 $\Omega(\epsilon^{-3})$ 下界。因此, 在通常的 ℓ_2 -几何中, SGD 是 minimax 最优的, 如果不改变几何结构、查询模型或驻点标准, 就不应期望最坏情况下的改进。

近期研究工作进一步探讨自适应方法或符号方法是否能在问题结构优化后, 在理论上优于 SGD。 [蒋等人 \[2025a\]](#) 表明, 在坐标平滑性和坐标噪声假设下, AdaGrad是可行的: 通过采用一种 ℓ_1 -平稳性度量, 他们证明了AdaGrad的上界和SGD的下界, 从而在维度因子上使 AdaGrad优于SGD。相应地, [克劳斯和刘 \[2025\]](#) 研究了自适应梯度算法在 (L_0, L_1) -平滑性下的下界, 表明AdaGrad的几个变体必然对松弛平滑性参数产生高阶依赖。

3 SignSGD: 上界和下界

在本节中, 我们首次对SignSGD的收敛性进行了精确描述。

3.1 符号和假设

我们用 $[T]$ 表示 $\{1, 2, \dots, T\}$, $|\mathbf{x}|$ 表示 $\mathbf{x}$ 的逐元素绝对值 $\in \mathbb{R}^d$ 。L加权向量范数对于 $\mathbf{L} \in \mathbb{R}_+^d$ 定义为 $\|\mathbf{x}\|_{\mathbf{L}}^2 := \mathbf{x}^\top \text{diag}(\mathbf{L})\mathbf{x}$ 。对于向量下降方法, 我们研究优化问题 $\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$, 其中 $f: \mathbb{R}^d \rightarrow \mathbb{R}$ 是可微分的。给定一个点 $\mathbf{x} \in \mathbb{R}^d$, 我们只能以噪声方式访问梯度 $\nabla f(\mathbf{x}) = [\nabla_1 f(\mathbf{x}) \dots, \nabla_d f(\mathbf{x})] \in \mathbb{R}^d$, 这将在假设2a中进一步定义。我们用 $\text{sign}(\cdot)$ 表示符号算子, 并且WLOG, 使用约定 $\text{sign}(0) = 0$, 这在我们的下界分析中可能会遇到。下面, 我们列出一些必要的假设。

假设1a (下界目标)。函数 f 有下界。存在 $f^* > -\infty$ 使得 $f(\mathbf{x}) \geq f^*$, 对所有 $\mathbf{x} \in \mathbb{R}^d$ 成立。我们进一步记 $\Delta = f(\mathbf{x}_0) - \inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$ 。

假设2a (可分离噪声模型)。在步骤 t 我们观察到一组相互独立的梯度 $g_t = \{\mathbf{g}_t^1, \dots, \mathbf{g}_t^B\}$ satisfying $\mathbb{E}[\mathbf{g}_t^b | \mathcal{F}_{t-1}] = \nabla f(\mathbf{x}_t) \forall b \in [B]$, 其中 $\mathcal{F}_t = \sigma(g_1, \dots, g_t)$ 表示自然 filtration。此外, 坐标方向的条件方差可分离地有界: 存在非负常数 $\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \dots, \sigma_d]$ 使得

$$\mathbb{E} \left[\left| \mathbf{g}_{t,i}^b - \nabla_i f(\mathbf{x}_t) \right|^2 \middle| \mathcal{F}_{t-1} \right] \leq \sigma_i^2, \quad \forall i \in [d], \forall b \in [B].$$

假设 1a 是随机非凸优化的标准且必要的 [[Arjevani 等人,2023](#)]。假设 2a 被广泛应用于 AdaGrad 等自适应方法的分析 [[Duchi 等人, 2011](#)], 以及 SignSGD 等符号方法 [[Bernstein 等人, 2018](#), [Lietal., 2025a](#), [Jiang 等人, 2025a](#), [Yu 等人, 2026b](#)]。
假设3a (ℓ_∞ -平滑)。目标函数 f 是 ℓ_∞ -平滑的, 如果存在非负常数 L_∞ , 使得对于所有 $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, 我们有

$$|f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle)| \leq \frac{L_\infty}{2} \|\mathbf{y} - \mathbf{x}\|_\infty^2.$$

Algorithm 1 SignSGD [Bernstein et al., 2018]	Algorithm 2 Muon [Jordan et al., 2024]
1: Require iteration number T , initial point $\mathbf{x}_0$, batch size B , learning rate η 2: for step $t = 0$ to $T - 1$ do 3: Compute stochastic gradient: $\mathbf{g}_t = \frac{1}{B} \sum_{b=1}^B \mathbf{g}_t^b$ 4: $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \text{sign}(\mathbf{g}_t)$ 5: end for	1: Require iteration number T , initial point $\mathbf{W}_0$, batch size B , learning rate η . 2: for step $t = 0$ to $T - 1$ do 3: Compute stochastic gradient: $\mathbf{G}_t = \frac{1}{B} \sum_{b=1}^B \mathbf{G}_t^b$ 4: $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \text{msign}(\mathbf{G}_t)$ 5: end for

Assumption 4a (Separable smoothness). The objective function f is $\mathbf{L}$ -separable smooth if there exists a non-negative vector $\mathbf{L} = [L_1, L_2, \dots, L_d]$ such that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we have

$$|f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle)| \leq \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_{\mathbf{L}}^2.$$

Separable smoothness (Assumption 4a) is widely used in analyses of SignSGD methods [Bernstein et al., 2018, 2019, Safaryan and Richtárik, 2021, Yu et al., 2026b], which naturally aligns with sign descent methods due to their separable nature. Balles et al. [2020] further discussed the relationship between Assumptions 3a and 4a, pointing out that the latter is a more generalized assumption. We present the following lemma to formalize this claim, whose proof can be found in Appendix A.

Lemma 1. If f is $\mathbf{L}$ -separable smooth, then it is also ℓ_∞ -smooth with $L_\infty = \|\mathbf{L}\|_1$.

3.2 Upper Bound Theory of SignSGD

For completeness, the whole algorithmic procedure of SignSGD is listed in Algorithm 1. Though the upper bounds of SignSGD and its variants under various settings have been widely studied [Bernstein et al., 2018, Karimireddy et al., 2019, Sun et al., 2023, Jiang et al., 2024, 2025b, Jiang and Zhang, 2025, Yu et al., 2026b], we present the upper bound analysis here for consistency of the paper.

Theorem 1. Run Algorithm 1 for T iterations under Assumptions 1a to 3a, by setting the hyperparameters as:

$$\eta = \sqrt{\frac{2\Delta}{L_\infty T}}, \quad B = \max \left\{ 1, \frac{\|\sigma\|_1^2}{\Delta L_\infty} T \right\}. \quad (1)$$

Denote $N = BT$ as the total complexity, Algorithm 1 guarantees:

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] = \mathcal{O} \left(\sqrt{\frac{L_\infty \Delta}{N}} + \left(\frac{\|\sigma\|_1^2 L_\infty \Delta}{N} \right)^{\frac{1}{4}} \right). \quad (2)$$

3.3 Lower Bound Theory of SignSGD

After establishing an upper bound for Algorithm 1, we present a corresponding lower bound under the same conditions, which immediately verifies the sharpness of the upper bound in Theorem 1.

Theorem 2. Fix $T \geq 1$ and a scaling parameter $\eta > 0$, and consider running Algorithm 1 for T iterations with batch size B . For any given parameters L_∞ , σ , and Δ , there exists a function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ and a stochastic gradient oracle such that:

1. f satisfies Assumptions 1a and 3a;
2. the stochastic gradients $\mathbf{g}_t$ satisfy Assumption 2a;
3. denote $N = BT$, the iterates generated by Algorithm 1 satisfy

$$\mathbb{E} \left[\min_{0 \leq t < T} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{L_\infty \Delta}{N}} + \left(\frac{\|\sigma\|_1^2 L_\infty \Delta}{N} \right)^{\frac{1}{4}} \right). \quad (3)$$

To the best of our knowledge, Theorem 2 is the first known tight lower bound for the SignSGD algorithm that does not depend on the dimension d explicitly. We briefly illustrate our proof techniques below, with the full analysis postponed to Appendix B.2.

算法1 SignSGD [Bernstein 等人, 2018]	算法2 Muon [Jordan 等人, 2024]
1: 要求 迭代次数 T , 初始点 $\mathbf{x}_0$, 批大小 B , 学习率 η 2: 对于 步骤 $t = 0$ 到 $T - 1$ 执行 3: 计算随机梯度: $\mathbf{g}_t = \frac{1}{B} \sum_{b=1}^B \mathbf{g}_t^b$ 4: $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \text{符号}(\mathbf{g}_t)$ 5: 结束for	1: 要求 迭代次数 T , 初始点 $\mathbf{W}_0$, 批大小 B , 学习率 η . 2: 对于 步长 $t = 0$ 到 $T - 1$ 执行 3: 计算随机梯度: $\mathbf{G}_t = \frac{1}{B} \sum_{b=1}^B \mathbf{G}_t^b$; $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \text{msign}(\mathbf{G}_t)$ 4: 5: 结束循环

假设4a (可分离平滑性). 目标函数 f 是 $\mathbf{L}$ -可分离平滑的, 如果存在一个非负向量 $\mathbf{L} = [L_1, L_2, \dots, L_d]$, 使得对于所有 $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, 我们有

$$|f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle)| \leq \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_{\mathbf{L}}^2.$$

可分离平滑性 (假设 4a) 在 SignSGD 方法的分析中广泛使用 [Bernstein 等人, 2018, 2019, Safaryan 和 Richtárik, 2021, Yu 等人, 2026b], 这自然地与符号下降方法一致, 因为它们具有可分离性. Balles 等人 [2020] 进一步讨论了假设 3a 和 4a 之间的关系, 指出后者是一个更通用的假设. 我们提出了以下引理来形式化这一主张, 其证明可以在附录 A 中找到.

引理 1. 如果 f 是 $\mathbf{L}$ -可分平滑的, 那么它也是 ℓ_∞ -平滑的, 复杂度为 $L_\infty = \|\mathbf{L}\|_1$.

3.2 SignSGD 的上界理论

为了完整性, SignSGD 的整个算法过程列在算法 1。尽管 SignSGD 及其在各种设置下的变体的上界已被广泛研究 [Bernstein 等人, 2018, Karimireddy 等人, 2019, Sun 等人, 2023, Jiang 等人, 2024, 2025b, Jiang 和 Zhang, 2025, Yu 等人, 2026b], 我们在这里给出上界分析以保持论文的一致性。

定理 1. 运行算法 1, 在假设 1a 到 3a 下运行 次迭代, 并将超参数设置为:

$$\eta = \sqrt{\frac{2\Delta}{L_\infty T}}, \quad B = \max \left\{ 1, \frac{\|\sigma\|_1^2}{\Delta L_\infty} T \right\}. \quad (1)$$

记 $N = BT$ 为总复杂度, 算法 1 保证:

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] = \mathcal{O} \left(\sqrt{\frac{L_\infty \Delta}{N}} + \left(\frac{\|\sigma\|_1^2 L_\infty \Delta}{N} \right)^{\frac{1}{4}} \right). \quad (2)$$

3.3 SignSGD 的下界理论

在建立算法 1 的上界后, 我们提出了相同条件下的下界, 这立即验证了定理 1 中上界的尖锐性。

定理 2. 固定 $T \geq 1$ 和缩放参数 $\eta > 0$, 并考虑运行算法 1 进行 T 次迭代, 批大小为 B 。对于任何给定的参数 L_∞ , σ , 和 Δ , 存在一个函数 $f: \mathbb{R}^d \rightarrow \mathbb{R}$ 和一个随机梯度预言机, 使得:

1. f 满足假设 1a 和 3a;
2. 随机梯度 $\mathbf{g}_t$ 满足假设 2a;
3. 记 $N = BT$, 算法 1 生成的迭代点满足

$$\mathbb{E} \left[\min_{0 \leq t < T} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{L_\infty \Delta}{N}} + \left(\frac{\|\sigma\|_1^2 L_\infty \Delta}{N} \right)^{\frac{1}{4}} \right). \quad (3)$$

据我们所知, 定理 2 是首个已知的、不依赖于维度的 SignSGD 算法的紧下界 d 。我们在此简要介绍我们的证明技术, 详细分析将推迟至附录 B.2。

Proof Sketch. To analyze the complexity of SignSGD, we divide the proof into five main steps.

Step 1: Dimensional Decomposition. We first reduce the d -dimensional optimization into d parallel one-dimensional problems by constructing a separable objective function $f(\mathbf{x}) = \sum_{i=1}^d p_i(\mathbf{x}_i)$, where $\mathbf{x}_i$ denotes the i -th coordinate. Utilizing Lemma 1, we can safely distribute the global ℓ_∞ -smoothness constant L_∞ and the suboptimality Δ across coordinates such that $\|\mathbf{L}\|_1 = L_\infty$ and $\sum_{i=1}^d \Delta_i = \Delta$.

Step 2: Constructing a 1D Resisting Oracle. We establish the worst-case lower bound for a single coordinate by constructing a “resisting oracle” [Nesterov et al., 2018]. Specifically, in the deterministic setting, we design a hard 1D function p_i that maintains a constant slope $p'_i(x) = -\epsilon$ across a sequence of N query points. Consequently, any algorithm that fails to escape this predefined region is strictly prevented from finding an ϵ -stationary point, as the gradient magnitude is uniformly bounded away from zero.

Step 3: Inducing Stall via Adversarial Bimodal Noise. To extend the 1D construction to the stochastic setting, we introduce an adversarial noise distribution. We define a specialized gradient oracle

$$\mathbb{P}(g_t^b = 0 \mid x_t) = \frac{\sigma_i^2}{\sigma_i^2 + \epsilon^2}, \quad \mathbb{P}\left(g_t^b = \frac{\sigma_i^2 + \epsilon^2}{\epsilon^2} p'_i(x_t) \mid x_t\right) = \frac{\epsilon^2}{\sigma_i^2 + \epsilon^2}. \quad (4)$$

This oracle is unbiased and has an exact variance of σ_i^2 . Crucially, this specific noise structure is designed to force the optimizer to stall, so that more total steps are needed to escape from previously defined “resisting oracle” when the variance σ_i^2 is large. We yield a rigorous 1D complexity lower bound under stochastic setting as:

$$\mathbb{E} \left[\min_t |p'(x_t)| \right] \geq C \max \left\{ \sqrt{\frac{L\Delta}{N}}, \left(\frac{L\Delta\sigma^2}{N} \right)^{\frac{1}{4}} \right\}.$$

Step 4: Dimensional Lifting and Adversarial Tuning. Finally, we aggregate the 1D lower bounds across all d dimensions. By adversarially tuning the coordinate-wise smoothness ($L_i \propto \sigma_i$) and the suboptimality (Δ_i) subject to their global constraints, we maximize the aggregated lower bound. This step successfully lifts the 1D result to the d -dimensional setting and recovers the exact dependency on $\|\sigma\|_1$ as stated in the theorem. $\square$

4 Why Sign Operator Works: Comparison Between SignSGD and SGD

In this section, we compare SignSGD with SGD from both theoretical and empirical perspectives. This comparison highlights the role of sign descent beyond its classical interpretation as a gradient-compression mechanism [Bernstein et al., 2018]: under Assumption 3a, the sign operator can also lead to provably improved complexity bounds.

4.1 Theoretical Study: SignSGD Converges Provably Faster than SGD

We introduce the following ℓ_1 -norm lower bound for SGD, whose proof can be found in Appendix C. This result characterizes the intrinsic hardness faced by SGD under our ℓ_∞ -smooth geometry, and therefore serves as the key benchmark for comparing against the upper bound of SignSGD.

Theorem 3 (SGD lower bound). Fix $T \geq 1$ and a scaling parameter $\eta > 0$, and consider running vanilla SGD for T iterations with batch size $B = 1$. For any given parameters L_∞ , σ , and Δ , there exists a function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ and a stochastic gradient oracle such that:

1. f satisfies Assumptions 1a and 3a;
2. the stochastic gradients $\mathbf{g}_t$ satisfy Assumption 2a;
3. the iterates generated by vanilla SGD satisfy

$$\mathbb{E} \left[\min_{t \in [T]} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{dL_\infty\Delta}{T}} + \left(\frac{d\|\sigma\|_2^2 L_\infty\Delta}{T} \right)^{\frac{1}{4}} \right). \quad (5)$$

Proof Sketch. We reduce the claim to the coordinate-wise lower bound of Jiang et al. [2025a]. Under Assumption 4a with smoothness vector $\mathbf{L} = (L_1, \dots, L_d)$ and separable variance vector $\sigma = (\sigma_1, \dots, \sigma_d)$, their result gives a hard instance for vanilla SGD as (24).

证明概要。为分析SignSGD的复杂度，我们将证明分为五个主要步骤。

步骤 1: 维度分解。 我们首先将 d 维优化问题转化为 d 一维并行问题，通过构造可分离的目标函数 $f(\mathbf{x}) = \sum_{i=1}^d p_i(\mathbf{x}_i)$ ，其中 $\mathbf{x}_i$ 表示第 i 个坐标。利用引理 1，我们可以安全地将全局 ℓ_∞ -光滑常数 L_∞ 和子最优性 Δ 分配到各个坐标上，使得 $\|\mathbf{L}\|_1 = L_\infty$ 和 $\sum_{i=1}^d \Delta_i = \Delta$ 。

步骤 2: 构建一维阻力算子。 我们通过构建一个“阻力算子” [Nesterov 等人, 2018]，来建立单个坐标的最坏情况下界。具体来说，在确定性设置下，我们设计一个保持恒定斜率的 p_i 困难一维函数 $p'_i(x) = -\epsilon$ ，在一系列 N 查询点中维持。因此，任何未能逃离预定义区域的算法都被严格阻止找到 ϵ -驻点，因为梯度幅度始终远离零有界。

步骤 3: 通过对抗双模态噪声诱导停滞。 为了将 1D 构造扩展到随机设置中，我们引入了一个对抗性噪声分布。我们定义了一个专门的梯度预言机

$$\mathbb{P}(g_t^b = 0 \mid x_t) = \frac{\sigma_i^2}{\sigma_i^2 + \epsilon^2}, \quad \mathbb{P}\left(g_t^b = \frac{\sigma_i^2 + \epsilon^2}{\epsilon^2} p'_i(x_t) \mid x_t\right) = \frac{\epsilon^2}{\sigma_i^2 + \epsilon^2}. \quad (4)$$

这个预言机是无偏的，其方差精确为 σ_i^2 。关键在于，这种特定的噪声结构被设计为迫使优化器停滞，以便在方差 σ_i^2 较大时需要更多总步数才能从先前定义的“抵抗预言机”中逃脱。我们在随机设置下得到一个严格的 1D 复杂度下界，如下：

$$\mathbb{E} \left[\min_t |p'(x_t)| \right] \geq C \max \left\{ \sqrt{\frac{L\Delta}{N}}, \left(\frac{L\Delta\sigma^2}{N} \right)^{\frac{1}{4}} \right\}.$$

步骤 4: 维度提升和对抗性调优。 最后，我们将所有 d 维度的 1D 下界进行聚合。通过对抗性地调优坐标方向的平滑度 ($L_i \propto \sigma_i$) 和次优性 (Δ_i) 并遵守它们的全局约束，我们最大化聚合的下界。这一步成功地将 1D 结果提升到 d 维设置中，并恢复了定理中声明的对 $\|\sigma\|_1$ 的精确依赖关系。 $\square$

4 为什么符号算子有效：SignSGD 与 SGD 的比较

在本节中，我们从理论和经验两个角度比较 SignSGD 与 SGD。这种比较突出了符号下降在梯度压缩机制这一经典解释之外的作用 [Bernstein 等人, 2018]: 在假设 3a 下，符号算子也可以导致可证明的复杂度下界提升。

4.1 理论研究：SignSGD 可证明地比 SGD 收敛更快

我们引入 SGD 的 ℓ_1 -范数下界，其证明可在附录 C 中找到。这一结果刻画了 SGD 在我们 ℓ_∞ -平滑几何结构下所面临的内在难度，因此成为与 SignSGD 上界进行比较的关键基准。

定理 3 (SGD 下界)。固定 $T \geq 1$ 和缩放参数 $\eta > 0$ ，考虑以批大小 $B = 1$ 运行 T 次迭代的原始 SGD。对于任何给定的参数 L_∞ 、 σ 和 Δ ，存在一个函数 $f: \mathbb{R}^d \rightarrow \mathbb{R}$ 和一个随机梯度预言机，使得：

1. f 满足假设 1a 和 3a；
2. 随机梯度 $\mathbf{g}_t$ 满足假设 2a；
3. vanillaSGD 生成的迭代满足

$$\mathbb{E} \left[\min_{t \in [T]} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{dL_\infty\Delta}{T}} + \left(\frac{d\|\sigma\|_2^2 L_\infty\Delta}{T} \right)^{\frac{1}{4}} \right). \quad (5)$$

证明思路。我们将该命题简化为Jiang等人关于坐标下界的结论 Jiang等人 [2025a]。在假设 4a 下，光滑度向量为 $\mathbf{L} = (L_1, \dots, L_d)$ ，且可分离方差向量为 $\sigma = (\sigma_1, \dots, \sigma_d)$ ，他们的结果为原始SGD提供了一个困难实例 (24)。

The key observation is that Lemma 1 converts any $\mathbf{L}$ -separable smooth function into an ℓ_∞ -smooth function with constant $\|\mathbf{L}\|_1$. Therefore, under the constraint $\|\mathbf{L}\|_1 = L_\infty$, the adversary is free to allocate the curvature across coordinates. Optimizing this allocation yields the desired lower bound under Assumption 3a.

For the deterministic term in (24), we choose a highly imbalanced smoothness vector, for example $L_1 \geq \|\mathbf{L}\|_1/2$ and $\|\mathbf{L}\|_1 = L_\infty$. Then $\|\mathbf{L}\|_\infty = \Omega(L_\infty)$, and the coordinate-wise lower bound gives

$$\mathbb{E} \left[\min_{t \in [T]} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{dL_\infty \Delta}{T}} \right). \quad (6)$$

This shows that, in the worst case, SGD suffers an additional factor d even in the noiseless part of the complexity.

For the stochastic term in (24), we allocate the curvature according to the noise profile by setting $L_i = (\sigma_i^2 / \|\sigma\|_2^2) L_\infty, i \in [d]$ so that $\sum_{i=1}^d L_i = L_\infty$ and it yields that

$$\mathbb{E} \left[\min_{t \in [T]} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\left(\frac{d \|\sigma\|_2^2 L_\infty \Delta}{T} \right)^{1/4} \right). \quad (7)$$

The two constructions above give two valid ℓ_∞ -smooth hard instances satisfying the same global parameters L_∞, σ, Δ . Taking the worse of them, and using $\max\{a, b\} = \Omega(a + b)$ for non-negative quantities a, b , we obtain the result in Theorem 3. $\square$

Comparison based on density function $\phi(\sigma)$. We proceed to compare the rate obtained in Theorem 1, which is the *upper bound* for SignSGD, and Theorem 3, which is the *lower bound* for SGD. We define the density function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ for noise σ as follows:

$$\phi(\sigma) = \frac{\|\sigma\|_1^2}{d \|\sigma\|_2^2} \in \left[\frac{1}{d}, 1 \right]. \quad (8)$$

The above function ϕ is a measure of how “dense” a vector is. $\phi(\sigma) \approx 1/d$ corresponds to highly skewed noise acting on only a few coordinates, whereas $\phi(\sigma) = 1$ represents perfectly uniform noise across all parameters. Now, we are able to compare SignSGD with SGD: to find an ϵ -stationary point in terms of the ℓ_1 -norm, the required complexity is

$$\Theta \left(\frac{L_\infty \Delta}{\epsilon^2} + \frac{\|\sigma\|_1^2 L_\infty \Delta}{\epsilon^4} \right) \quad \text{for SignSGD}, \quad (9)$$

$$\text{and } \Omega \left(\frac{dL_\infty \Delta}{\epsilon^2} + \frac{\|\sigma\|_1^2 L_\infty \Delta}{\phi(\sigma) \epsilon^4} \right) \quad \text{for SGD}. \quad (10)$$

In view of the above two complexity bounds, we make the following observations.

1. **Deterministic term.** For the first noiseless term, SignSGD reduces the complexity of SGD by a factor of d , revealing its superior dependence on the dimensional factors.
2. **Stochastic term.** Since $1/d \leq \phi(\sigma) \leq 1$, we underline that the second noise-dependent term of SignSGD is still *strictly better* than SGD. Moreover, when the distribution of the noise σ is sparse, SignSGD further reduces the required iteration count of SGD by a factor of d .

As shown above, our results provide the first problem setting in which SignSGD achieves a provably better complexity bound (in terms of the dimensional dependence) than SGD.

4.2 Empirical Study

In previous sections, we have discussed how SignSGD can potentially outperform SGD, especially under the ℓ_∞ -smooth and sparse noise settings. To verify the theoretical results, we conduct a comprehensive empirical study for SignSGD and SGD under a variety of regimes. Our code is available at https://github.com/Dingzhen230/SignSGD_Outperforms_SGD.

关键观察是引理1 将任何 $\mathbf{L}$ -可分离光滑函数转化为 ℓ_∞ -光滑函数，其常数为 $\|\mathbf{L}\|_1$ 。因此，在约束 $\|\mathbf{L}\|_1 = L_\infty$ 下，对手可以自由分配曲率到各个坐标。优化这种分配即可在假设3a下得到所需的下界。

对于 (24) 中的确定性项，我们选择一个高度不平衡的光滑度向量，例如 $L_1 \geq \|\mathbf{L}\|_1/2$ 和 $\|\mathbf{L}\|_1 = L_\infty$ 然后 $\|\mathbf{L}\|_\infty = \Omega(L_\infty)$ ，并且坐标意义上的下界给出

$$\mathbb{E} \left[\min_{t \in [T]} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{dL_\infty \Delta}{T}} \right). \quad (6)$$

这表明，在最坏情况下，即使是在复杂度的无噪声部分，SGD也会遭受一个额外的因子 d 。

对于 (24) 中的随机项，我们根据噪声分布分配曲率，通过设置 $L_i = (\sigma_i^2 / \|\sigma\|_2^2) L_\infty, i \in [d]$ 使得 $\sum_{i=1}^d L_i = L_\infty$ 并且它得到

$$\mathbb{E} \left[\min_{t \in [T]} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\left(\frac{d \|\sigma\|_2^2 L_\infty \Delta}{T} \right)^{1/4} \right). \quad (7)$$

上述两种构造给出了两个有效的 ℓ_∞ -光滑硬实例，满足相同的全局参数 L_∞, σ, Δ 。取其中较差的一个，并使用 $\max\{a, b\} = \Omega(a + b)$ 对于非负量 a, b ，我们得到定理 3 的结果。 $\square$

基于密度函数的比较 $\phi(\sigma)$ 。 我们继续比较定理1中得到的速率，该速率是SignSGD的上界，以及定理3中得到的下界。我们定义密度函数 $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ 对于噪声 σ 如下：

$$\phi(\sigma) = \frac{\|\sigma\|_1^2}{d \|\sigma\|_2^2} \in \left[\frac{1}{d}, 1 \right]. \quad (8)$$

上述函数 ϕ 是衡量向量“稠密”程度的一个指标。 $\phi(\sigma) \approx 1/d$ 对应于仅作用于少数坐标的高偏斜噪声，而 $\phi(\sigma) = 1$ 则表示在所有参数上完全均匀的噪声。现在，我们能够比较 SignSGD与SGD：要在 ℓ_1 -范数下找到一个 ϵ -定点，所需的复杂度是

$$\Theta \left(\frac{L_\infty \Delta}{\epsilon^2} + \frac{\|\sigma\|_1^2 L_\infty \Delta}{\epsilon^4} \right) \quad \text{for SignSGD}, \quad (9)$$

$$\text{and } \Omega \left(\frac{dL_\infty \Delta}{\epsilon^2} + \frac{\|\sigma\|_1^2 L_\infty \Delta}{\phi(\sigma) \epsilon^4} \right) \quad \text{for SGD}. \quad (10)$$

鉴于上述两个复杂度界限，我们做出以下观察。

1. **确定性项。** 对于第一个无噪声项，SignSGD将SGD的复杂度降低了 d 倍，揭示了其对维度因子的优越依赖性。
2. **随机项。** 由于 $1/d \leq \phi(\sigma) \leq 1$ ，我们强调SignSGD的第二项噪声相关项仍然严格优于SGD。此外，当噪声 σ 的分布是稀疏时，SignSGD进一步将SGD所需的迭代次数降低了 d 倍。

如上所述，我们的结果提供了首个问题设置，其中 SignSGD 在复杂度界限（以维度依赖性衡量）上实现了可证明的优于 SGD 的表现。

4.2 实证研究

在之前的章节中，我们讨论了 SignSGD 如何有可能优于 SGD，尤其是在 ℓ_∞ -光滑和稀疏噪声设置下。为了验证理论结果，我们对 SignSGD 和 SGD 在各种情况下进行了全面的实证研究。我们的代码可在 https://github.com/Dingzhen230/SignSGD_优于_SGD。

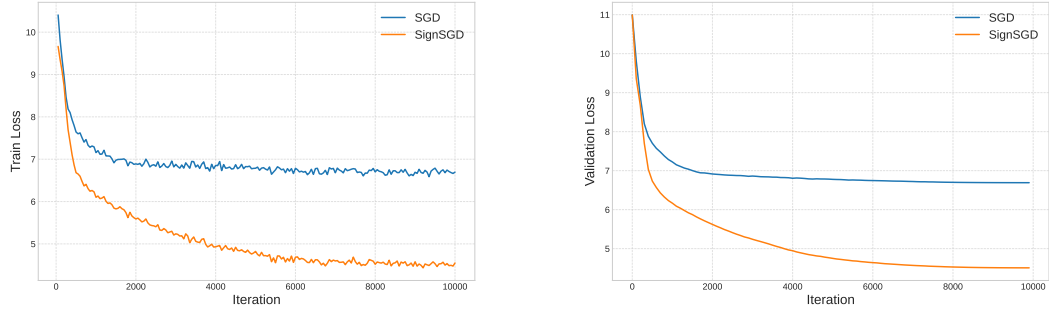


Figure 1: The training and validation loss for nanoGPT trained on C4.

4.2.1 Experiments on Numerical Examples

To amplify the difference between the two optimizers, we manually crafted two numerical example functions that fit into the extreme cases stated in Section 4.1. Due to limited space, the experimental results of these numerical examples are shown in Figure 3. Our findings are summarized as follows.

1. **Deterministic example.** SignSGD outperforms SGD when facing a ℓ_∞ -geometry target and a sufficiently large d . Therefore, we set our target function as an imbalanced quadratic objective $f(\mathbf{x}) = 1/2 \sum_{i=1}^d L_i x_i^2$ for $\mathbf{x} \in \mathbb{R}^d$. To better demonstrate the difference, we set $d = 5000$ and $L_1 = 1000, L_i = 1, i \geq 2$. From Figure 3a we can observe that SGD performs worse than SignSGD because it is heavily penalized for the 1st coordinate where L_1 dominates L_∞ .
2. **Stochastic example.** SignSGD outperforms SGD when facing sparse noise structure. We switch to a simple quadratic objective function $f(x) = \|\mathbf{x}\|_2^2/2$ for $\mathbf{x} \in \mathbb{R}^{100}$. Then we add Gaussian noise $\mathcal{N}(0, 100^2)$ to only the first component of the gradient, which further results in maximizing the sparsity of σ as $\phi(\sigma) = 1/d$ under such a setting. Results in Figure 3b align with our findings, as SignSGD consistently outperforms SGD facing sparse noise.

4.2.2 Experiments on nanoGPT Pretraining

Following the setup detailed in Appendix E, we train the nanoGPT model [Karpathy, 2022] on the C4 dataset [Raffel et al., 2020] using SGD and SignSGD [Bernstein et al., 2018]. The training and validation learning curves are shown in Figure 1. We observe a substantial performance gap between the two optimizers: while SGD achieves a rapid decrease in loss during the very early stages of training, its progress drastically slows down and plateaus shortly after. In contrast, SignSGD maintains a consistent and significantly faster convergence rate throughout the entire pretraining process.

To understand the fundamental cause of this severe performance degradation in SGD, and to verify whether it stems from the highly sparse noise distribution identified in our theory, we leverage the empirical framework in Bernstein et al. [2018] to investigate the noise distributions in practice. Specifically, we track the dynamic evolution of the gradient noise density function ($\phi(\sigma)$) during training and compare the optimization trajectory of our LLM against a standard CNN baseline (ResNet-20 [He et al., 2016] on CIFAR-10 [Krizhevsky et al., 2009]). The comparative results are presented in Figure 2.

As illustrated in Figure 2, the noise distribution trends between CNN and LLM training exhibit a striking discrepancy. In the CNN training, the gradient noise starts relatively sparse but rapidly homogenizes during the early epochs, eventually stabilizing into a highly dense state ($\phi \approx 0.7$). This aligns perfectly with classical assumptions where noise become more Gaussian-like over time, as the expected density ϕ for an isotropic Gaussian distribution analytically converges to $2/\pi \approx 0.637$. Conversely, the LLM starts with a distribution with $\phi \approx 0.3$ and becomes *increasingly sparse* as training progresses, plunging to extreme levels of $\phi \rightarrow 0.03$ before the middle of training procedure.

These empirical findings perfectly corroborate our theoretical insights by providing a unified explanation for the contrasting optimization dynamics observed across different modalities. In CNNs, where the gradient noise rapidly homogenizes into a dense state as visualized in Figure 2b, our theory correctly predicts that SignSGD and SGD should perform comparably, as seen in Bernstein et al. [2018]. Conversely, in LLM pretraining, Figure 2a shows that the noise distribution becomes

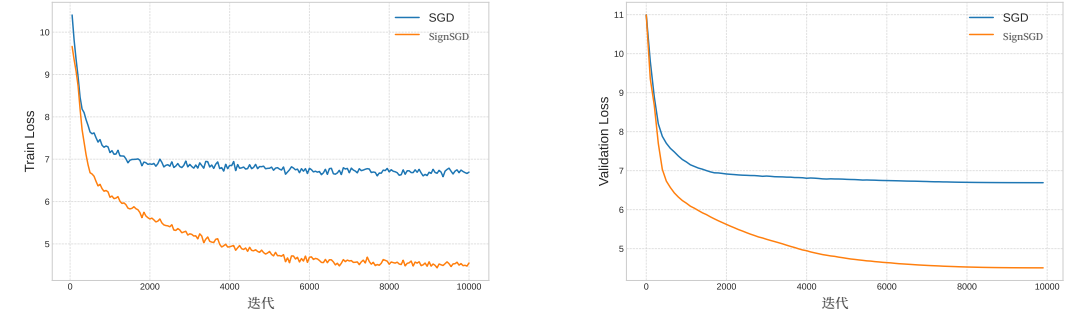


图1：在C4上训练的nanoGPT的训练和验证损失。

4.2.1 数值示例实验

为了放大两种优化器之间的差异，我们手动设计了两个符合第 4.1 节所述极端情况的数值示例函数。由于篇幅限制，这些数值示例的实验结果如图 3 所示。我们的发现总结如下。

1. **确定性示例。** 当面对 ℓ_∞ -几何目标并存在足够大的 d 时，SignSGD 会优于 SGD。因此，我们设置目标函数为不平衡二次目标 $f(\mathbf{x}) = 1/2 \sum_{i=1}^d L_i x_i^2$ ，其中 $\mathbf{x} \in \mathbb{R}^d$ 。为了更好地展示差异，我们设置 $d = 5000$ 和 $L_1 = 1000, L_i = 1, i \geq 2$ 。从图 3a 中可以观察到，由于 SGD 在第1坐标上受到严重惩罚，其中 L_1 占主导地位 L_∞ ，因此 SGD 的表现不如 SignSGD。
2. **随机示例。** 当面对稀疏噪声结构时，SignSGD 会优于 SGD。我们切换到简单的二次目标函数 $f(x) = \|\mathbf{x}\|_2^2/2$ ，其中 $\mathbf{x} \in \mathbb{R}^{100}$ 。然后我们对梯度仅第一分量添加高斯噪声 $\mathcal{N}(0, 100^2)$ ，这进一步导致 σ 的稀疏性最大化，在当前设置下 $\phi(\sigma) = 1/d$ 。图 3b 中的结果与我们的发现一致，即面对稀疏噪声时，SignSGD 始终优于 SGD。

4.2.2 nanoGPT 预训练实验

根据附录 E 中详细设置的说明，我们训练了 nanoGPT 模型 [Karpathy, 2022] 在 C4 数据集 [Raffel 等人, 2020] 上使用 SGD 和 SignSGD [Bernstein 等人, 2018]。训练和验证学习曲线显示在图 1 中。我们观察到两种优化器之间存在显著的性能差距：虽然 SGD 在训练的非常早期阶段实现了损失的快速下降，但其进展在不久之后急剧减慢并进入平台期。相比之下，SignSGD 在整个预训练过程中始终保持一致且显著更快的收敛速度。

为了理解 SGD 中这种严重的性能退化根本原因，并验证其是否源于我们在理论中识别的高度稀疏噪声分布，我们利用 Bernstein 等人 [2018] 中的经验框架来实际研究噪声分布。具体而言，我们追踪训练过程中梯度噪声密度函数 ($\phi(\sigma)$) 的动态演变，并将我们的 LLM 优化轨迹与标准 CNN 基线（ResNet-20 [He 等人, 2016] 在 CIFAR-10 [Krizhevsky 等人, 2009]）进行比较。比较结果展示在图 2 中。

如图 2 所示，CNN 和 LLM 训练中的噪声分布趋势存在显著差异。在 CNN 训练中，梯度噪声最初相对稀疏，但在早期轮次中迅速同质化，最终稳定为一个高度密集的状态 ($\phi \approx 0.7$)。这与经典假设相符，即噪声随时间推移变得更接近高斯分布，因为各向同性高斯分布的预期密度 ϕ 通过解析收敛到 $2/\pi \approx 0.637$ 。相反，LLM 以 $\phi \approx 0.3$ 的分布开始，并在训练过程中变得越来越稀疏，在训练过程的中期之前降至极低的水平。

这些实证发现完美地证实了我们的理论见解，它们为不同模态中观察到的对比优化动态提供了一个统一的解释。在 CNN 中，梯度噪声如图 2b 所示迅速同质化成密集状态，我们的理论正确预测 SignSGD 和 SGD 的表现应该相当，正如 Bernstein 等人所看到的。相反，在 LLM 预训练中，图 2a 显示噪声分布变得 2a 越来越稀疏且严重偏斜。标准 SGD，其更新与梯度幅度成正比，被这种极端的方差异质性所困。然而，基于符号的优化器通过利用对梯度幅度不变的逐坐标更新（如图 1 所示产生了决定性的性能优势），安全地应对这种极端稀疏性。我们的理论框架成功解释了 CNN 中的性能相当性和 LLM 中 SignSGD 的显著优越性，证实了基于符号的方法更适合 LLM，正如第 4.1 节所确立的。

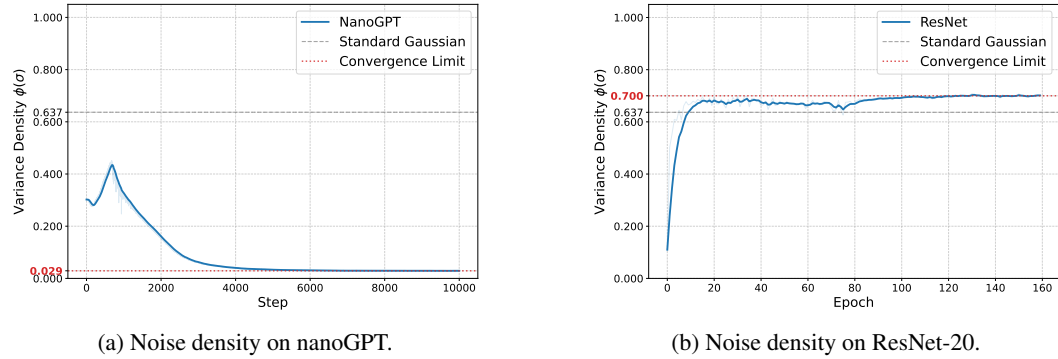


Figure 2: Evolution of gradient noise sparsity density $\phi(\sigma)$ along the training trajectory of SGD. Figure 2a shows noise density of LLM(nanoGPT-124m) tracked continuously across 10000 iterations. Figure 2b shows noise density of CNN (ResNet-20) tracked continuously across 160 epochs.

increasingly sparse and heavily skewed. Standard SGD, whose updates are proportional to gradient magnitudes, is paralyzed by this extreme variance heterogeneity. Sign-based optimizers, however, safely navigate this extreme sparsity by utilizing coordinate-wise updates that are invariant to gradient magnitudes (yielding the decisive performance advantage seen in Figure 1). Our theoretical framework successfully reconciles both the performance parity in CNNs and the significant superiority of SignSGD in LLMs, confirming that sign-based methods are intrinsically more suited for LLMs as established in Section 4.1.

5 Matrix Optimizers

While vector-based sign methods effectively handle separable scaling, modern LLM architectures rely heavily on matrix multiplications. This motivates extending our framework to the matrix domain. The Muon algorithm is introduced by Jordan et al. [2024], and there lies essential similarity between SignSGD and Muon: Bernstein and Newhouse [2024] pointed out that SignSGD can be regarded as *steepest descent under infinity norm*, while Muon can be regarded as *steepest descent under spectral norm*. Now that we have already illustrated the edge of SignSGD in Section 4, we will show in the sequel that it’s natural to extend our previous framework from vectors to the matrix domain.

5.1 Notations and Assumptions

We denote the set of $m \times m$ positive semi-definite (PSD) matrices by $\mathbb{S}^m$. For any $\mathbf{X} \in \mathbb{R}^{m \times n}$. The matrix sign operator is defined as $\text{msign}(\mathbf{X}) := \mathbf{U}\mathbf{V}^\top$, where $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ is the (compact) singular value decomposition (SVD) of $\mathbf{X} \in \mathbb{R}^{m \times n}$. Following common practice [Li and Hong, 2025, Shen et al., 2025, Sato et al., 2025, Chang et al., 2025, Pan et al., 2025], we assume zero numerical error in Newton–Schulz algorithm. The inner product between matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{m \times n}$ is denoted by $\langle \mathbf{Y}, \mathbf{X} \rangle := \text{Tr}(\mathbf{Y}^\top \mathbf{X})$. We let $\|\cdot\|_{\text{op}}$, $\|\cdot\|_*$, and $\|\cdot\|_F$ denote the matrix operator norm, nuclear norm, and Frobenius norm, respectively. For a vector $\mathbf{v} \in \mathbb{R}^m$, $\text{diag}(\mathbf{v}) \in \mathbb{R}^{m \times m}$ denotes the rectangular diagonal matrix whose elements are given by $(\text{diag}(\mathbf{v}))_{ij} = v_i$ if $i = j$ and 0 otherwise. Throughout this section, we consider the matrix optimization problem $\min_{\mathbb{R}^{m \times n}} F(\mathbf{W})$ under the following assumptions.

Assumption 1b (Lower bounded objective). The function $F \in \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is bounded from below. There exists $F^* > -\infty$ such that $F(\mathbf{W}) \geq F^*$ holds for all $\mathbf{W} \in \mathbb{R}^{m \times n}$. We further denote $\Delta = F(\mathbf{W}_0) - \inf_{\mathbf{W} \in \mathbb{R}^{m \times n}} F(\mathbf{W})$.

Assumption 2b (Extension of Assumption 2a in matrix form). At step t we observe a mini-batch of mutually independent gradients $G_t = \{\mathbf{G}_t^1, \dots, \mathbf{G}_t^B\}$ satisfying $\mathbb{E}[\mathbf{G}_t^b | \mathcal{F}_{t-1}] = \nabla f(\mathbf{W}_t), \forall b \in [B]$ where $\mathcal{F}_t = \sigma(G_1, \dots, G_t)$ denotes the natural filtration. Denote by $\mathbf{N}_t^b = \mathbf{G}_t^b - \nabla f(\mathbf{W}_t)$, there exists $\mathbf{\Sigma} \in \mathbb{S}^m$ such that

$$\mathbb{E}[(\mathbf{N}_t^b)(\mathbf{N}_t^b)^\top | \mathcal{F}_{t-1}] \preceq \mathbf{\Sigma}^2, \quad \forall b \in [B].$$

Assumption 3b (Spectral norm smoothness). We say $F: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is L_* -spectral norm smooth if for any $\mathbf{W}, \mathbf{W}' \in \mathbb{R}^{m \times n}$, it holds that

$$|F(\mathbf{W}') - (F(\mathbf{W}) + \langle \nabla F(\mathbf{W}), \mathbf{W}' - \mathbf{W} \rangle)| \leq \frac{L_*}{2} \|\mathbf{W}' - \mathbf{W}\|_{\text{op}}^2.$$

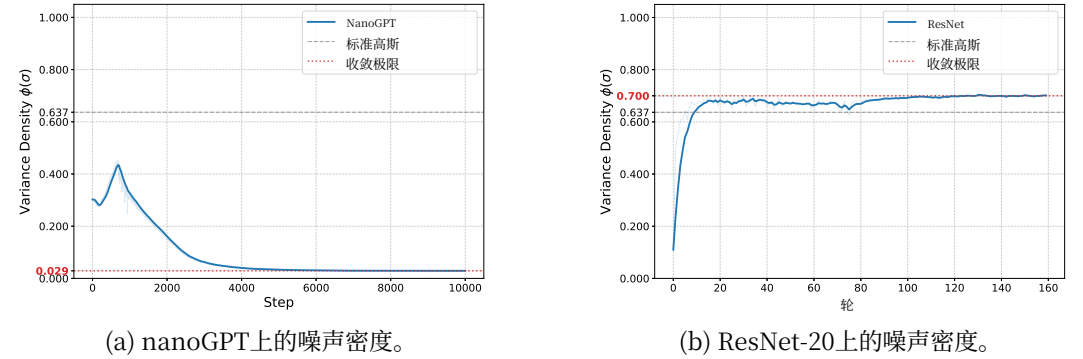


图2: 梯度噪声稀疏密度沿SGD训练轨迹的演化 $\phi(\sigma)$ 图2a显示了LLM(nanoGPT-124m)在10000次迭代中持续追踪的噪声密度。图2b显示了CNN (ResNet-20)在160轮中持续追踪的噪声密度。

这些实证发现完美地证实了我们的理论见解，它们为不同模式中观察到的对比优化动态提供了一个统一的解释。在CNN中，梯度噪声如图2b所示迅速同质化成密集状态，我们的理论正确预测SignSGD和SGD的表现应该相当，正如Bernstein等人所看到的。相反，在LLM预训练中，图2a显示噪声分布变得2a越来越稀疏且严重偏斜。标准SGD，其更新与梯度幅度成正比，被这种极端的方差异质性所困。然而，基于符号的优化器通过利用对梯度幅度不变的逐坐标更新（如图1所示产生了决定性的性能优势），安全地应对这种极端稀疏性。我们的理论框架成功解释了CNN中的性能相当性和LLM中SignSGD的显著优越性，证实了基于符号的方法更适合LLM，正如第4.1节所确立的。

5 矩阵优化器

虽然基于向量的符号方法能有效处理可分离缩放，但现代LLM架构严重依赖矩阵乘法。这促使我们将我们的框架扩展到矩阵领域。Muon算法由Jordan等人 [2024], 引入，SignSGD与Muon之间存在本质相似性：Bernstein和Newhouse [2024] 指出SignSGD可以被视为无穷范数下的最速下降，而Muon可以被视为谱范数下的最速下降。既然我们在第4节已经说明了SignSGD的优势，我们将在后续展示将我们之前的框架从向量扩展到矩阵领域是自然的。

5.1 符号和假设

我们用 $m \times m$ 表示正定 (PSD) 矩阵的集合 $\mathbb{S}^m$ 。对于任何 $\mathbf{X} \in \mathbb{R}^{m \times n}$ 。矩阵符号算子定义为 $\text{msign}(\mathbf{X}) := \mathbf{U}\mathbf{V}^\top$ ，其中 $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ 是 $\mathbf{X} \in \mathbb{R}^{m \times n}$ 的 (紧) 奇异值分解 (SVD)。遵循 Li 和 Hong, 2025、Shen 等人, 2025、Sato 等人, 2025、Chang 等人, 2025、Pan 等人 [2025], 的惯例，我们假设牛顿-舒尔茨算法中没有数值误差。矩阵 $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{m \times n}$ 之间的内积用 $\langle \mathbf{Y}, \mathbf{X} \rangle := \text{Tr}(\mathbf{Y}^\top \mathbf{X})$ 表示。我们分别用 $\|\cdot\|_{\text{op}}$ 、 $\|\cdot\|_*$ 和 $\|\cdot\|_F$ 表示矩阵算子范数、核范数和弗罗贝尼乌斯范数。对于向量 $\mathbf{v} \in \mathbb{R}^m$ ， $\text{diag}(\mathbf{v}) \in \mathbb{R}^{m \times m}$ 表示其元素为 $(\text{diag}(\mathbf{v}))_{ij} = v_i$ (当 $i = j$ 为真时) 和 0 (否则) 的矩形对角矩阵。在本节中，我们在以下假设下考虑矩阵优化问题 $\min_{\mathbb{R}^{m \times n}} F(\mathbf{W})$ 。

假设2b (下界目标)。函数 $F \in \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ 有下界。存在 $F^* > -\infty$ 使得 $F(\mathbf{W}) \geq F^*$ 对所有 $\mathbf{W} \in \mathbb{R}^{m \times n}$ 成立。我们进一步记 $\Delta = F(\mathbf{W}_0) - \inf_{\mathbf{W} \in \mathbb{R}^{m \times n}} F(\mathbf{W})$ 。

假设2b (矩阵形式下假设2a的推广)。在步骤 t 我们观察到一组相互独立的梯度 $G_t = \{G_t^1, \dots, G_t^B\}$ 满足 $\mathbb{E}[G_t^b | \mathcal{F}_{t-1}] = \nabla f(W_t) \forall b \in [B]$ ，其中 $\mathcal{F}_t = \sigma(G_1, \dots, G_t)$ 表示自然filtration。记 $\mathbf{N}_t^b = G_t^b - \nabla f(W_t)$ ，存在 $\mathbf{\Sigma} \in \mathbb{S}^m$ 使得

$$\mathbb{E}[(\mathbf{N}_t^b)(\mathbf{N}_t^b)^\top | \mathcal{F}_{t-1}] \preceq \mathbf{\Sigma}^2, \quad \forall b \in [B].$$

假设3b (谱范数平滑性)。我们说 $F: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ 是 L_* -谱范数平滑的，如果对于任何 $\mathbf{W}, \mathbf{W}' \in \mathbb{R}^{m \times n}$ ，它都满足

$$|F(\mathbf{W}') - (F(\mathbf{W}) + \langle \nabla F(\mathbf{W}), \mathbf{W}' - \mathbf{W} \rangle)| \leq \frac{L_*}{2} \|\mathbf{W}' - \mathbf{W}\|_{\text{op}}^2.$$

Assumption 2b can be viewed as an extension of Assumption 2a into matrix space, and has been widely used for analyzing Muon and other matrix Optimizers (An et al. [2025, Assumption 3], Pan et al. [2025, Assumption 3]). Assumption 3b accounts for the distinct structure of matrix parameters, which can be viewed as an extension of Assumption 3a into matrix space.

5.2 Upper bound Theory of Muon

The Muon algorithm is presented in Algorithm 2. We state its upper bound below, with the proof deferred to Appendix D.1.

Theorem 4 (Muon upper bound). Run Algorithm 2 for T iterations under Assumptions 1b to 3b, by setting the hyperparameters as:

$$\eta = \sqrt{\frac{2\Delta}{L_*T}}, \quad B = \max \left\{ 1, \frac{\|\Sigma\|_*^2}{\Delta L_*} T \right\}. \quad (11)$$

Denote $N = BT$ as the total complexity, Algorithm 2 guarantees:

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla F(\mathbf{W}_t)\|_* \right] = \mathcal{O} \left(\sqrt{\frac{L_*\Delta}{N}} + \left(\frac{\|\Sigma\|_*^2 L_*\Delta}{N} \right)^{1/4} \right). \quad (12)$$

A key feature of Theorem 4 is that the bound does not depend explicitly on the matrix dimension $\min\{m, n\}$, in sharp contrast to existing analyses of Muon that incur such dimension dependence [Li and Hong, 2025, Shen et al., 2025, Chang et al., 2025, Huang et al., 2025].

5.3 Lower bound Theory of Muon

After establishing the upper bound for Algorithm 2, we present a corresponding lower bound under the same conditions, which verifies the sharpness of Theorem 4. The proof is deferred to Appendix D.2.

Theorem 5 (Muon lower bound). Fix $T \geq 1$ and a scaling parameter $\eta > 0$, and consider running Algorithm 2 for T iterations with batch size B . For any given parameters L_∞ , σ , and Δ , there exists a function $f: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ and a stochastic gradient oracle such that:

1. f satisfies Assumptions 1b and 3b;
2. the stochastic gradients $\mathbf{G}_t$ satisfy Assumption 2b;
3. denote $N = BT$, the iterates generated by Algorithm 2 satisfy

$$\mathbb{E} \left[\min_t \|\nabla F(\mathbf{W}_t)\|_* \right] = \mathcal{O} \left(\sqrt{\frac{L_*\Delta}{N}} + \left(\frac{\|\Sigma\|_*^2 L_*\Delta}{N} \right)^{1/4} \right). \quad (13)$$

To our best knowledge, Theorem 5 is the first lower complexity bound for Muon. Existing stochastic lower bounds, such as Arjevani et al. [2023], are formulated for vector-valued methods under Euclidean geometry, and thus do not directly apply to spectral-norm smoothness and nuclear-norm stationarity. Our theorem fills this gap and shows that the dimension-free upper bound in Theorem 4 is unimprovable in our matrix geometry. We outline the proof below.

Proof Sketch. Our proof establishes a strict geometric and dynamic equivalence, mathematically reducing the Muon optimization in the matrix domain to the SignSGD optimization in the vector domain. We outline the proof in the following five steps.

Step 1: Constructing the Matrix Objective with Orthogonal Alignments. We lift the hard vector instance $f(\mathbf{x})$ into the matrix domain by defining $F(\mathbf{W}) = \sum_{i=1}^m f_i((\mathbf{Q}\mathbf{W}\mathbf{P}^\top)_{ii})$. The reasons why we additionally introduce two orthogonal matrices $\mathbf{Q}$ and $\mathbf{P}$ here are:

- $\mathbf{Q}$ is explicitly used to align the separable vector noise with the target matrix covariance Σ ;
- $\mathbf{P}$ is used as the projection matrix, projecting the $m \times n$ matrix into an $m \times m$ subspace.

Step 2: Designing the Structured Matrix Oracle. To mirror the stochasticity, we construct a structured stochastic matrix gradient $\mathbf{G}_t = \mathbf{Q}^\top \text{diag}(\mathbf{g}_t)\mathbf{P}$. Driven by the alignment in Step 1, this specific design perfectly satisfies the required matrix noise covariance $\Sigma = \mathbf{Q}^\top \text{diag}(\sigma)\mathbf{Q}$, while safely isolating the worst-case vector noise $\mathbf{g}_i$ strictly within the projected subspace.

假设 2b 可以看作是假设 2a 在矩阵空间中的扩展，并且已被广泛用于分析 Muon 和其他矩阵优化器 (An et al. [2025, 假设 3], Pan et al. [2025, 假设 3])。假设 3b 考虑了矩阵参数的独特结构，可以看作是假设 3a 在矩阵空间中的扩展。

5.2 Muon 上界理论

Muon 算法在算法 2 中给出。我们在此陈述其上界，证明则推迟至附录 D.1。Theorem 4 (Muon 上界)。

运行算法 2，在假设 1b 至 3b 的条件下，进行 T 次迭代的参数设置为：

$$\eta = \sqrt{\frac{2\Delta}{L_*T}}, \quad B = \max \left\{ 1, \frac{\|\Sigma\|_*^2}{\Delta L_*} T \right\}. \quad (11)$$

记 $N = BT$ 为总复杂度，算法 2 保证：

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla F(\mathbf{W}_t)\|_* \right] = \mathcal{O} \left(\sqrt{\frac{L_*\Delta}{N}} + \left(\frac{\|\Sigma\|_*^2 L_*\Delta}{N} \right)^{1/4} \right). \quad (12)$$

定理 4 的一个关键特征是，该界限不显式依赖于矩阵维度，与现有对 Muon 的分析形成鲜明对比，后者存在这种维度依赖性 [李和黄, 2025, 沈等人, 2025, 张等人, 2025, 黄等人, 2025]。

5.3 Muon 的下界理论

在建立了算法 2 的上界后，我们在相同条件下提出了相应的下界，验证了定理 4 的尖锐性。证明推迟到附录 D.2。

定理 5 (Muon 下界) (Muon 下界)。固定 $T \geq 1$ 和缩放参数 $\eta > 0$ ，考虑运行算法 2，迭代次数为 T ，批大小为 B 。对于任何给定的参数 L_∞ ， σ ，和 Δ ，存在一个函数 $f: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ 和一个随机梯度预言机，使得：

1. f 满足假设 1b 和 3b；
2. 随机梯度 $\mathbf{G}_t$ 满足假设 2b；
3. 记 $N = BT$ ，算法 2 生成的迭代满足

$$\mathbb{E} \left[\min_t \|\nabla F(\mathbf{W}_t)\|_* \right] = \mathcal{O} \left(\sqrt{\frac{L_*\Delta}{N}} + \left(\frac{\|\Sigma\|_*^2 L_*\Delta}{N} \right)^{1/4} \right). \quad (13)$$

据我们所知，定理 5 是首个针对 Muon 的下复杂度界。现有的随机下界，如 Arjevani 等人 [2023]，是为欧几里得几何下的向量值方法制定的，因此不直接适用于谱范数平滑性和核范数平稳性。我们的定理填补了这一空白，并表明在矩阵几何中，定理 4 中的无维上界是不可改进的。我们将在下文概述证明。

证明概要。 我们的证明建立了一种严格的几何和动态等价关系，将矩阵域中的 Muon 优化数学地还原为向量域中的 SignSGD 优化。我们将在以下五个步骤中概述证明。

步骤 1: 构造具有正交对齐的矩阵目标函数。 我们将硬向量实例 $f(\mathbf{x})$ 提升到矩阵域，通过定义 $F(\mathbf{W}) = \sum_{i=1}^m f_i((\mathbf{Q}\mathbf{W}\mathbf{P}^\top)_{ii})$ 。我们在这里额外引入两个正交矩阵 $\mathbf{Q}$ 和 $\mathbf{P}$ 的原因如下：

- $\mathbf{Q}$ 被明确用于将可分离向量噪声与目标矩阵协方差 Σ 对齐；
- $\mathbf{P}$ 被用作投影矩阵，将 $m \times n$ 矩阵投影到 $m \times m$ 子空间中。

步骤 2: 设计结构化矩阵 Oracle。 为了模拟随机性，我们构造了一个结构化随机矩阵梯度 $\mathbf{G}_t = \mathbf{Q}^\top \text{diag}(\mathbf{g}_t)\mathbf{P}$ 。受步骤 1 中对齐的驱动，这种特定设计完美满足所需的矩阵噪声协方差 $\Sigma = \mathbf{Q}^\top \text{diag}(\sigma)\mathbf{Q}$ ，同时安全地将最坏情况向量噪声 $\mathbf{g}_i$ 严格隔离在投影子空间内。

Step 3: Establishing Trajectory Equivalence via SVD. Since $\mathbf{Q}^\top$ and $\mathbf{P}$ are naturally orthogonal matrices, they seamlessly form the left and right singular vectors of $\mathbf{G}_t$ without altering the SVD transformation. Consequently, the SVD orthogonalization bypasses them and explicitly acts as a scalar $\text{sign}(\cdot)$ operator on the inner diagonal entries. Muon’s matrix update $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \text{msign}(\mathbf{G}_t)$ mathematically collapses into the exact SignSGD vector update: $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \text{sign}(\mathbf{g}_t)$.

Step 4: Aligning Geometric Constraints. We establish a rigorous one-to-one mapping between the governing metrics of both domains. We prove that the vector ℓ_∞ -smoothness flawlessly translates to the matrix spectral norm smoothness ($L_* = L_\infty$), and the trace norm of the matrix noise covariance matches the ℓ_1 -norm of the vector noise ($\|\Sigma\|_* = \|\sigma\|_1$).

Step 5: Transferring the Complexity Bound. Finally, because orthogonal transformations preserve singular values, we show that the trace norm (nuclear norm) of the matrix gradient $\|\nabla F(\mathbf{W}_t)\|_*$ is precisely equivalent to the ℓ_1 -norm of the vector gradient $\|\nabla f(\mathbf{x}_t)\|_1$, enabling us to directly invoke the vector bounds from Theorem 2 to establish the exact $\Omega(\cdot)$ lower bound for Muon. $\square$

6 Conclusion

In this work, we bridge the longstanding gap between the empirical superiority of sign-based optimizers and their theoretical guarantees in non-convex stochastic optimization. By moving from the standard ℓ_2 framework to an ℓ_∞ -smooth, coordinate-wise noise setting with ℓ_1 -stationarity, we give an optimal complexity-theoretic characterization of SignSGD and identify a problem geometry in which it provably outperforms SGD. Specifically, we prove matching upper and lower bounds for SignSGD, establish a strictly worse lower bound for SGD, and show that the resulting separation becomes most pronounced under sparse or highly heterogeneous noise. We further extend this framework to matrix optimization and obtain matching bounds for Muon under spectral norm smoothness and nuclear norm stationarity. Our empirical results corroborate these theoretical findings. Both controlled sparse noise toy problems and GPT-2 pretraining exhibit the skewed gradient noise structure leveraged by our theory, supporting the viewpoint that sign-based methods enjoy a genuine geometric advantage in real-world high-dimensional problems, such as LLM pretraining.

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步骤 3：通过 SVD 建立轨迹等价性由于 $\mathbf{Q}^\top$ 和 $\mathbf{P}$ 是天然的正交矩阵，它们无缝地形成 $\mathbf{G}_t$ 的左奇异向量和右奇异向量，而不会改变 SVD 变换。因此，SVD 正交化绕过它们，并明确地作为标量符号 ($\cdot$) 运算符作用于对角线内部元素。Muon 的矩阵更新 $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \text{msign}(\mathbf{G}_t)$ 数学上简化为精确的 SignSGD 向量更新： $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \text{sign}(\mathbf{g}_t)$ 。

步骤 4：对齐几何约束。 我们建立了两个域的支配度量之间严格的一一映射。我们证明，向量 ℓ_∞ -平滑性完美地转换为矩阵谱范数平滑性 ($L_* = L_\infty$)，矩阵噪声协方差的迹范数与向量噪声的 ℓ_1 -范数 ($\|\Sigma\|_* = \|\sigma\|_1$) 相匹配。

步骤 5：转移复杂度界限。 最后，因为正交变换保持奇异值，我们证明矩阵梯度的迹范数（核范数） $\|\nabla F(\mathbf{W}_t)\|_*$ 精确地等同于 ℓ_1 -范数的向量梯度 $\|\nabla f(\mathbf{x}_t)\|_1$ ，使我们能够直接引用定理2中的向量界限来建立精确的 $\Omega(\cdot)$ 下界用于 Muon。 $\square$

6 结论

在这项工作中，我们弥合了基于符号的优化器经验优越性与非凸随机优化中的理论保证之间的长期差距。通过从标准的 ℓ_2 框架转向 ℓ_∞ -平滑、逐坐标噪声设置并具有 ℓ_1 -平稳性，我们给出了 SignSGD 的最优复杂度理论特征，并确定了一种问题几何结构，其中 SignSGD 可证明优于 SGD。具体来说，我们证明了 SignSGD 的匹配上下界，为 SGD 建立了严格更差的下界，并表明在稀疏或高度异构噪声下，由此产生的分离最为明显。我们进一步将此框架扩展到矩阵优化，并在谱范数平滑性和核范数平稳性下为 Muon 获得了匹配界限。我们的经验结果证实了这些理论发现。在受控的稀疏噪声玩具问题和 GPT-2 预训练中，都表现出我们理论利用的倾斜梯度噪声结构，支持基于符号的方法在高维现实问题（如 LLM 预训练）中享有真正的几何优势的观点。

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A Proof of Lemma 1

Suppose f is $\mathbf{L}$ -separable smooth, we have

$$\begin{aligned} |f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle)| &\leq \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_{\mathbf{L}}^2 = \frac{1}{2} \sum_{i=1}^d L_i (\mathbf{y}_i - \mathbf{x}_i)^2 \\ &\leq \frac{1}{2} \sum_{i=1}^d L_i \|\mathbf{y} - \mathbf{x}\|_{\infty}^2 = \frac{\|\mathbf{L}\|_1}{2} \|\mathbf{y} - \mathbf{x}\|_{\infty}^2. \end{aligned}$$

Therefore f also satisfies Assumption 3a with $L_{\infty} = \|\mathbf{L}\|_1$.

B Analysis for SignSGD

B.1 Upper Bound for SignSGD

Proof of Theorem 1. Under Assumption 3a, we have:

$$\begin{aligned} f(\mathbf{x}_{t+1}) &\leq f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + \frac{L_{\infty}}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|_{\infty}^2 \\ &= f(\mathbf{x}_t) - \langle \nabla f(\mathbf{x}_t), \eta \text{sign}(\mathbf{g}_t) \rangle + \frac{\eta^2 L_{\infty}}{2} \\ &= f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \eta(\text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign}(\mathbf{g}_t)) \rangle \\ &\quad - \langle \nabla f(\mathbf{x}_t), \eta \text{sign}(\nabla f(\mathbf{x}_t)) \rangle + \frac{\eta^2 L_{\infty}}{2} \\ &= f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \eta(\text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign}(\mathbf{g}_t)) \rangle - \eta \|\nabla f(\mathbf{x}_t)\|_1 + \frac{\eta^2 L_{\infty}}{2} \\ &\leq f(\mathbf{x}_t) + 2\eta \|\nabla f(\mathbf{x}_t) - \mathbf{g}_t\|_1 - \eta \|\nabla f(\mathbf{x}_t)\|_1 + \frac{\eta^2 L_{\infty}}{2}, \end{aligned}$$

where the last inequality is due to

$$\begin{aligned} &\langle \nabla f(\mathbf{x}_t), \eta(\text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign}(\mathbf{g}_t)) \rangle \\ &= \sum_{i=1}^d \nabla_i f(\mathbf{x}_t) * \eta(\text{sign}(\nabla_i f(\mathbf{x}_t)) - \text{sign}(\mathbf{g}_{t,i})) \\ &\leq \sum_{i=1}^d 2\eta |\nabla_i f(\mathbf{x}_t)| * \mathbb{I}(\text{sign}(\nabla_i f(\mathbf{x}_t)) \neq \text{sign}(\mathbf{g}_{t,i})) \\ &\leq 2\eta \sum_{i=1}^d |\nabla_i f(\mathbf{x}_t) - \mathbf{g}_{t,i}| * \mathbb{I}(\text{sign}(\nabla_i f(\mathbf{x}_t)) \neq \text{sign}(\mathbf{g}_{t,i})) \\ &\leq 2\eta \sum_{i=1}^d |\nabla_i f(\mathbf{x}_t) - \mathbf{g}_{t,i}| = 2\eta \|\nabla f(\mathbf{x}_t) - \mathbf{g}_t\|_1. \end{aligned}$$

Rearranging the obtained relation and summing up yields

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \frac{\Delta}{\eta T} + 2\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{g}_t\|_1 \right] + \frac{\eta L_{\infty}}{2}. \quad (14)$$

we denote $\xi_t = \nabla f(\mathbf{x}_t) - \mathbf{g}_t$ as the noise of gradient in iteration t . Due to i.i.d characteristic of ξ_t , we have:

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{g}_t\|_1 \right] &= \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\xi_t\|_1 \right] = \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\xi_t\|_1] \\ &= \frac{1}{T} \sum_{t=1}^T \sum_{i=1}^d \mathbb{E} [|\xi_{t,i}|] \leq \frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{B}}, \end{aligned} \quad (15)$$

一个引理的证明1

假设 f 是 $\mathbf{L}$ -可分平滑的, 我们有

$$\begin{aligned} |f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle)| &\leq \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_{\mathbf{L}}^2 = \frac{1}{2} \sum_{i=1}^d L_i (\mathbf{y}_i - \mathbf{x}_i)^2 \\ &\leq \frac{1}{2} \sum_{i=1}^d L_i \|\mathbf{y} - \mathbf{x}\|_{\infty}^2 = \frac{\|\mathbf{L}\|_1}{2} \|\mathbf{y} - \mathbf{x}\|_{\infty}^2. \end{aligned}$$

因此 f 也满足假设3a, 并且 $L_{\infty} = \|\mathbf{L}\|_1$.

B SignSGD分析

B.1 SignSGD的上界

定理1的证明。在假设3a下, 我们有:

$$\begin{aligned} f(\mathbf{x}_{t+1}) &\leq f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + \frac{L_{\infty}}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|_{\infty}^2 \\ &= f(\mathbf{x}_t) - \langle \nabla f(\mathbf{x}_t), \eta \text{sign}(\mathbf{g}_t) \rangle + \frac{\eta^2 L_{\infty}}{2} \\ &= f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \eta(\text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign}(\mathbf{g}_t)) \rangle \\ &\quad - \langle \nabla f(\mathbf{x}_t), \eta \text{sign}(\nabla f(\mathbf{x}_t)) \rangle + \frac{\eta^2 L_{\infty}}{2} \\ &= f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \eta(\text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign}(\mathbf{g}_t)) \rangle - \eta \|\nabla f(\mathbf{x}_t)\|_1 + \frac{\eta^2 L_{\infty}}{2} \\ &\leq f(\mathbf{x}_t) + 2\eta \|\nabla f(\mathbf{x}_t) - \mathbf{g}_t\|_1 - \eta \|\nabla f(\mathbf{x}_t)\|_1 + \frac{\eta^2 L_{\infty}}{2}, \end{aligned}$$

其中最后一个不等式是由于

$$\begin{aligned} &\langle \nabla f(\mathbf{x}_t), \eta(\text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign}(\mathbf{g}_t)) \rangle \\ &= \sum_{i=1}^d \nabla_i f(\mathbf{x}_t) * \eta(\text{sign}(\nabla_i f(\mathbf{x}_t)) - \text{sign}(\mathbf{g}_{t,i})) \\ &\leq \sum_{i=1}^d 2\eta |\nabla_i f(\mathbf{x}_t)| * \mathbb{I}(\text{sign}(\nabla_i f(\mathbf{x}_t)) \neq \text{sign}(\mathbf{g}_{t,i})) \\ &\leq 2\eta \sum_{i=1}^d |\nabla_i f(\mathbf{x}_t) - \mathbf{g}_{t,i}| * \mathbb{I}(\text{sign}(\nabla_i f(\mathbf{x}_t)) \neq \text{sign}(\mathbf{g}_{t,i})) \\ &\leq 2\eta \sum_{i=1}^d |\nabla_i f(\mathbf{x}_t) - \mathbf{g}_{t,i}| = 2\eta \|\nabla f(\mathbf{x}_t) - \mathbf{g}_t\|_1. \end{aligned}$$

重新排列所得关系式并求和得到

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \frac{\Delta}{\eta T} + 2\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{g}_t\|_1 \right] + \frac{\eta L_{\infty}}{2}. \quad (14)$$

我们记 $\xi_t = \nabla f(\mathbf{x}_t) - \mathbf{g}_t$ 为迭代 t 中的梯度噪声。由于 ξ_t 的独立同分布特性, 我们有:

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{g}_t\|_1 \right] &= \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\xi_t\|_1 \right] = \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\xi_t\|_1] \\ &= \frac{1}{T} \sum_{t=1}^T \sum_{i=1}^d \mathbb{E} [|\xi_{t,i}|] \leq \frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{B}}, \end{aligned} \quad (15)$$

where the inequality follows from Jensen's inequality for the variance of the mini-batch mean. Putting (15) back to (14), and setting $\eta = \sqrt{2\Delta/L_\infty T}$, We get:

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \sqrt{\frac{2\Delta L_\infty}{T}} + 2 \frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{B}}. \quad (16)$$

If the first term dominates, which means

$$\sqrt{\frac{2\Delta L_\infty}{T}} \geq 2\|\boldsymbol{\sigma}\|_1 \geq 2 \frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{B}}, \quad (17)$$

by setting $B = 1$, we have

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \sqrt{\frac{2\Delta L_\infty}{T}} + 2 \frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{B}} \leq (2 + \sqrt{2}) \sqrt{\frac{\Delta L_\infty}{N}}.$$

Otherwise, we set $B = \|\boldsymbol{\sigma}\|_1^2 / \Delta L_\infty T$, which implies

$$\sqrt{\frac{2\Delta L_\infty}{T}} = \left(\frac{4\|\boldsymbol{\sigma}\|_1^2 \Delta L_\infty}{N} \right)^{\frac{1}{4}}. \quad (18)$$

Plugging (18) into (16), we get

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \left(\frac{4\|\boldsymbol{\sigma}\|_1^2 \Delta L_\infty}{N} \right)^{\frac{1}{4}} + 2 \left(\frac{\|\boldsymbol{\sigma}\|_1^2 \Delta L_\infty}{N} \right)^{\frac{1}{4}}. \quad (19)$$

Combining the above two cases together, we have

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] = \mathcal{O} \left(\sqrt{\frac{\Delta L_\infty}{N}} + \frac{(\|\boldsymbol{\sigma}\|_1)^{\frac{1}{2}} (\Delta L_\infty)^{\frac{1}{4}}}{(N)^{\frac{1}{4}}} \right).$$

This completes the proof. $\square$

B.2 Lower Bound for SignSGD

To exploit the separable nature of SignSGD before generalizing to d dimensions, we start by bounding the complexity of Algorithm 1 in finding stationary points in one dimension:

Lemma 2. For any positive integer n , suppose that ϵ satisfies

$$\epsilon \leq \frac{1}{2\sqrt{n}} \quad (20)$$

Let $x_1 \in \mathbb{R}$ and $x_t = x_1 + (t-1)\eta$ for any $2 \leq t \leq n+1$. Then there exists a function $p: \mathbb{R} \rightarrow \mathbb{R}$ such that: (i) p has a 1-Lipschitz gradient; (ii) $p(x_1) - \inf p \leq 1$ (iii) $p'(x_t) = -\epsilon$ for any $t \in [1, n]$.

Proof. We construct the function p as:

$$p(x) = \begin{cases} -\epsilon(x - x_1) & x \in (-\infty, x_1]; \\ \phi_{t,\epsilon}(x) + c_t & x \in (x_t, x_{t+1}]; \\ \frac{1}{2}(x - x_{t+1})^2 - \epsilon(x - x_{t+1}) + c_{N+1} & x \in (x_{n+1}, \infty), \end{cases}$$

where the values $\{c_t\}_{t=1}^n$ are chosen to ensure that the function p is continuous. The function $\phi_{t,\epsilon}(x)$ is defined as below:

$$\phi_{t,\epsilon}(x) = \begin{cases} \frac{1}{2}(x - x_t)^2 - \epsilon(x - x_t) & x \in \left(x_t, \frac{x_t + x_{t+1}}{2}\right]; \\ -\frac{1}{2}(x - x_{t+1})^2 - \epsilon(x - x_{t+1}) & x \in \left(\frac{x_t + x_{t+1}}{2}, x_{t+1}\right]. \\ + \frac{(x_{t+1} - x_t)^2}{4} - (x_{t+1} - x_t)\epsilon & \end{cases}$$

其中不等式源自于小批量均值的方差所满足的 Jensen 不等式。将 (15) 代回 (14), 并设 $\eta = \sqrt{2\Delta/L_\infty T}$, 我们得到:

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \sqrt{\frac{2\Delta L_\infty}{T}} + 2 \frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{B}}. \quad (16)$$

如果第一项占主导地位, 这意味着

$$\sqrt{\frac{2\Delta L_\infty}{T}} \geq 2\|\boldsymbol{\sigma}\|_1 \geq 2 \frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{B}}, \quad (17)$$

通过设置 $B = 1$, 我们得到

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \sqrt{\frac{2\Delta L_\infty}{T}} + 2 \frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{B}} \leq (2 + \sqrt{2}) \sqrt{\frac{\Delta L_\infty}{N}}.$$

否则, 我们设置 $B = \|\boldsymbol{\sigma}\|_1^2 / \Delta L_\infty T$, 这意味着

$$\sqrt{\frac{2\Delta L_\infty}{T}} = \left(\frac{4\|\boldsymbol{\sigma}\|_1^2 \Delta L_\infty}{N} \right)^{\frac{1}{4}}. \quad (18)$$

将 (18) 代入 (16), 我们得到

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \left(\frac{4\|\boldsymbol{\sigma}\|_1^2 \Delta L_\infty}{N} \right)^{\frac{1}{4}} + 2 \left(\frac{\|\boldsymbol{\sigma}\|_1^2 \Delta L_\infty}{N} \right)^{\frac{1}{4}}. \quad (19)$$

将上述两种情况结合起来, 我们得到

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] = \mathcal{O} \left(\sqrt{\frac{\Delta L_\infty}{N}} + \frac{(\|\boldsymbol{\sigma}\|_1)^{\frac{1}{2}} (\Delta L_\infty)^{\frac{1}{4}}}{(N)^{\frac{1}{4}}} \right).$$

这完成了证明。 $\square$

B.2 SignSGD的下界

为了利用SignSGD的可分离性并在推广到 d 维之前, 我们首先通过在单维中寻找驻点的算法复杂度来界定:

引理2. 对于任何正整数 n , 假设 ϵ 满足

$$\epsilon \leq \frac{1}{2\sqrt{n}} \quad (20)$$

设 $x_1 \in \mathbb{R}$ 和 $x_t = x_1 + (t-1)\eta$ 对任何 $2 \leq t \leq n+1$ 成立。那么存在一个函数 $p: \mathbb{R} \rightarrow \mathbb{R}$ 使得: (i) p 具有1-Lipschitz梯度; (ii) $p(x_1) - \inf p \leq 1$ (iii) $p'(x_t) = -\epsilon$ 对任何 $t \in [1, n]$ 成立。

证明。我们构造函数 p 为:

$$p(x) = \begin{cases} -\epsilon(x - x_1) & x \in (-\infty, x_1]; \\ \phi_{t,\epsilon}(x) + c_t & x \in (x_t, x_{t+1}]; \\ \frac{1}{2}(x - x_{t+1})^2 - \epsilon(x - x_{t+1}) + c_{N+1} & x \in (x_{n+1}, \infty), \end{cases}$$

在 $\{c_t\}_{t=1}^n$ 处, 值被选择以确保函数 p 是连续的。函数 $\phi_{t,\epsilon}(x)$ 定义如下:

$$\phi_{t,\epsilon}(x) = \begin{cases} \frac{1}{2}(x - x_t)^2 - \epsilon(x - x_t) & x \in \left(x_t, \frac{x_t + x_{t+1}}{2}\right]; \\ -\frac{1}{2}(x - x_{t+1})^2 - \epsilon(x - x_{t+1}) & x \in \left(\frac{x_t + x_{t+1}}{2}, x_{t+1}\right]. \\ + \frac{(x_{t+1} - x_t)^2}{4} - (x_{t+1} - x_t)\epsilon & \end{cases}$$

Without loss of generality we can assume $x_1 = \eta$, otherwise we just need to perform a translation to our function p . Plugging in $x_t = t\eta$, we have:

$$\phi_{t,\epsilon}(x) = \begin{cases} \frac{1}{2}(x - t\eta)^2 - \epsilon(x - t\eta) & x \in \left(x_t, \frac{x_t + x_{t+1}}{2}\right]; \\ -\frac{1}{2}(x - (t+1)\eta)^2 - \epsilon(x - (t+1)\eta) + \frac{\eta^2}{4} - \eta\epsilon & x \in \left(\frac{x_t + x_{t+1}}{2}, x_{t+1}\right]. \end{cases}$$

Further, we can give the value c_t as follows:

$$c_1 = 0, \quad c_{t+1} = t \left(\frac{1}{4}\eta^2 - \eta\epsilon \right). \quad (21)$$

From the definition of p , we have

$$p(x) \geq \begin{cases} 0, & x \in (-\infty, x_1]; \\ \min \left\{ c_t - \frac{1}{2}\epsilon^2, c_{t+1} \right\}, & x \in (x_t, x_{t+1}]; \\ c_{n+1} - \frac{1}{2}\epsilon^2, & x \in (x_{n+1}, +\infty), \end{cases}$$

which can be written as

$$\begin{aligned} \inf p &\geq \min_t c_t - \frac{1}{2}\epsilon^2 \\ &= \min \left\{ 0, n \left(\frac{1}{4}\eta^2 - \eta\epsilon \right) \right\} - \frac{1}{2}\epsilon^2. \end{aligned}$$

This implies that

$$\begin{aligned} p(\mathbf{x}_1) - \inf p &\leq \max \left\{ \frac{1}{2}\epsilon^2, n \left(\eta\epsilon - \frac{1}{4}\eta^2 \right) + \frac{1}{2}\epsilon^2 \right\} \\ &\leq \max \left\{ \frac{1}{2}\epsilon^2, n\epsilon^2 + \frac{1}{2}\epsilon^2 \right\} \\ &= n\epsilon^2 + \frac{1}{2}\epsilon^2. \end{aligned}$$

With (20), we have

$$n\epsilon^2 + \frac{1}{2}\epsilon^2 \leq \frac{1}{4} + \frac{1}{8n} < 1.$$

This completes the proof. $\square$

Lemma 3. Consider running **SignSGD** on a one-dimensional smooth function p with the scaling parameter η and batch size B . For any $L > 0$ and $\Delta > 0$, there exists a function $p : \mathbb{R} \rightarrow \mathbb{R}$ and a corresponding stochastic gradient oracle g_t such that: (i) p has L -Lipschitz gradients and $p(x_1) - \inf p \leq \Delta$ (ii) the stochastic gradient g_t^b is unbiased and has a bounded variance of σ^2 (iii) Given ϵ such that $\epsilon \leq \sqrt{L\Delta}$, if $N = BT \leq L\Delta(4\epsilon^2 + \sigma^2)/128\epsilon^4$, then we have $\mathbb{E}[\min |p'(x_i)|] \geq \epsilon$. Which further implies that:

$$\mathbb{E} \left[\min_t |p'(x_t)| \right] \geq C \max \left\{ \left(\frac{L\Delta\sigma^2}{N} \right)^{\frac{1}{4}}, \sqrt{\frac{L\Delta}{N}} \right\}.$$

Proof of Lemma 3. We set $x_1 = \eta$. Without loss of generality, we can assume that $L = 1$ and $\Delta = 1$.

Now we define $N := 1/16\epsilon^2$, which surely satisfies the requirement in (20) with 2ϵ . According to Lemma 2, there exists a function $p : \mathbb{R} \rightarrow \mathbb{R}$ such that (i) its gradient is 1-Lipschitz; (ii) $p(x_1) - \inf p \leq 1$; (iii) $p'(x_t) = -2\epsilon$ for any $1 \leq t \leq N + 1$.

Consider running **SignSGD** on function p with the stochastic gradient oracle g_t^b given as below:

$$\mathbb{P}(g_t^b = 0 \mid x_t) = \frac{\sigma^2}{\sigma^2 + 4\epsilon^2}, \mathbb{P}\left(g_t^b = \frac{\sigma^2 + 4\epsilon^2}{4\epsilon^2} p'(x_t) \mid x_t\right) = \frac{4\epsilon^2}{\sigma^2 + 4\epsilon^2}. \quad b \in [B]$$

不失一般性, 我们可以假设 $x_1 = \eta$, 否则我们只需要对函数 p 进行平移。代入 $x_t = t\eta$, 我们得到:

$$\phi_{t,\epsilon}(x) = \begin{cases} \frac{1}{2}(x - t\eta)^2 - \epsilon(x - t\eta) & x \in \left(x_t, \frac{x_t + x_{t+1}}{2}\right]; \\ -\frac{1}{2}(x - (t+1)\eta)^2 - \epsilon(x - (t+1)\eta) + \frac{\eta^2}{4} - \eta\epsilon & x \in \left(\frac{x_t + x_{t+1}}{2}, x_{t+1}\right]. \end{cases}$$

此外, 我们可以给出 c_t 的值如下:

$$c_1 = 0, \quad c_{t+1} = t \left(\frac{1}{4}\eta^2 - \eta\epsilon \right). \quad (21)$$

根据 p 的定义, 我们有

$$p(x) \geq \begin{cases} 0, & x \in (-\infty, x_1]; \\ \min \left\{ c_t - \frac{1}{2}\epsilon^2, c_{t+1} \right\}, & x \in (x_t, x_{t+1}]; \\ c_{n+1} - \frac{1}{2}\epsilon^2, & x \in (x_{n+1}, +\infty), \end{cases}$$

这可以写成

$$\begin{aligned} \inf p &\geq \min_t c_t - \frac{1}{2}\epsilon^2 \\ &= \min \left\{ 0, n \left(\frac{1}{4}\eta^2 - \eta\epsilon \right) \right\} - \frac{1}{2}\epsilon^2. \end{aligned}$$

这意味着

$$\begin{aligned} p(\mathbf{x}_1) - \inf p &\leq \max \left\{ \frac{1}{2}\epsilon^2, n \left(\eta\epsilon - \frac{1}{4}\eta^2 \right) + \frac{1}{2}\epsilon^2 \right\} \\ &\leq \max \left\{ \frac{1}{2}\epsilon^2, n\epsilon^2 + \frac{1}{2}\epsilon^2 \right\} \\ &= n\epsilon^2 + \frac{1}{2}\epsilon^2. \end{aligned}$$

使用 (20), 我们得到

$$n\epsilon^2 + \frac{1}{2}\epsilon^2 \leq \frac{1}{4} + \frac{1}{8n} < 1.$$

这完成了证明。 $\square$

引理 3. 考虑在单变量光滑函数 p 上运行 SignSGD, 其缩放参数为 η , 批大小为 B 。对于任意 $L > 0$ 和 $\Delta > 0$, 存在一个函数 $p : \mathbb{R} \rightarrow \mathbb{R}$ 以及一个相应的随机梯度预言机 g_t , 使得: (i) p 具有 L -Lipschitz梯度和 $p(x_1) - \inf p \leq \Delta$ (ii) 随机梯度 g_t^b 无偏且具有有界方差 σ^2 (iii) 给定 ϵ 使得 $\epsilon \leq \sqrt{L\Delta}$, 如果 $N = BT \leq L\Delta(4\epsilon^2 + \sigma^2)/128\epsilon^4$, 则我们有 $\mathbb{E}[\min |p'(x_i)|] \geq \epsilon$ 。这进一步意味着:

$$\mathbb{E} \left[\min_t |p'(x_t)| \right] \geq C \max \left\{ \left(\frac{L\Delta\sigma^2}{N} \right)^{\frac{1}{4}}, \sqrt{\frac{L\Delta}{N}} \right\}.$$

引理 3的证明。我们设 $x_1 = \eta$ 。不失一般性, 我们可以假设 $L = 1$ 和 $\Delta = 1$ 。

现在我们定义 $N := 1/16\epsilon^2$, 这显然满足(20)中的要求, 其中 2ϵ 。根据引理 2, 存在一个函数 $p : \mathbb{R} \rightarrow \mathbb{R}$, 使得 (i) 其梯度是1-Lipschitz; (ii) $p(x_1) - \inf p \leq 1$; (iii) $p'(x_t) = -2\epsilon$ 对于任何 $1 \leq t \leq N + 1$ 。

考虑运行 **SignSGD** 在函数 p 上, 并使用随机梯度预言机 g_t^b 如下所示:

$$\mathbb{P}(g_t^b = 0 \mid x_t) = \frac{\sigma^2}{\sigma^2 + 4\epsilon^2}, \mathbb{P}\left(g_t^b = \frac{\sigma^2 + 4\epsilon^2}{4\epsilon^2} p'(x_t) \mid x_t\right) = \frac{4\epsilon^2}{\sigma^2 + 4\epsilon^2}. \quad b \in [B]$$

It is straightforward to verify that g_t^b satisfies the requirement.

Now we denote m_t as the number of steps that g_t is non-zero during the first t steps with $m_0 = 0$, which means that for m_t steps the x_t goes forward by η and $t - m_t$ steps the algorithm does nothing to x_t since $g_t = 0, \text{sign}(g_t) = 0$. We denote $M = m_T$. By definition, we have

$$\mathbb{E}(M) = T \times \left(1 - \left(1 - \frac{4\epsilon^2}{\sigma^2 + 4\epsilon^2}\right)^B\right) \leq T \times \left(1 - 1 + \frac{4B\epsilon^2}{\sigma^2 + 4\epsilon^2}\right) = \frac{4\epsilon^2}{\sigma^2 + 4\epsilon^2}BT.$$

According to Markov's inequality we have

$$\mathbb{P}(M \geq 2\mathbb{E}[M]) \leq \frac{1}{2},$$

which means that with probability at least $\frac{1}{2}$, we have $M \leq 2\mathbb{E}[M] \leq n$. Moreover, we can use induction to prove that $x_t = \eta + \eta M_{t-1}$, which means that $p'(x_t) = -2\epsilon$. Finally, we can bound

$$\mathbb{E} \left[\min_t |p'(x_t)| \right] \geq \frac{1}{2} \mathbb{E} \left[\min_t |p'(x_t)| \mid M \leq n \right] = \epsilon.$$

Since the equation above holds for any $N = TB \leq L\Delta(4\epsilon^2 + \sigma^2)/128\epsilon^4$, by setting $\epsilon = \sqrt{L\Delta/32N}$, we get

$$N = \frac{L\Delta}{32\epsilon^2} \leq L\Delta \frac{4\epsilon^2 + \sigma^2}{128\epsilon^4}. \quad (22)$$

Similarly we can set $\epsilon = (L\Delta\sigma^2/128N)^{1/4}$ to satisfy the condition. Therefore for some constant C , we have

$$\mathbb{E} \left[\min_t |p'(x_t)| \right] \geq C \max \left\{ \left(\frac{L\Delta\sigma^2}{N} \right)^{\frac{1}{4}}, \sqrt{\frac{L\Delta}{N}} \right\}. \quad (23)$$

This completes the proof. $\square$

Lemma 3 states the complexity lower bound for Algorithm 1 for a one-dimensional function. Now we can extend to a dimension d result, which will yield the bound in Theorem 2.

Proof of Theorem 2. First by applying Lemma 3 to each coordinate, we have

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] \geq \sum_{i=1}^d \max \left\{ \left(\frac{L_i \Delta_i \sigma_i^2}{N} \right)^{\frac{1}{4}}, \sqrt{\frac{L_i \Delta_i}{N}} \right\}.$$

By choosing $\Delta_i = L_i \Delta / \|L\|_1$, we have:

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] \geq \sum_{i=1}^d \sqrt{\frac{L_i \Delta_i}{N}} = \sum_{i=1}^d L_i \sqrt{\frac{\Delta}{\|L\|_1 N}} = \sqrt{\frac{\|L\|_1 \Delta}{N}}.$$

Then, by choosing $\Delta_i = \frac{\sigma_i^{\frac{2}{3}} L_i^{\frac{1}{3}}}{\sum_{j=1}^d \sigma_j^{\frac{2}{3}} L_j^{\frac{1}{3}}} \Delta$, we get

$$\begin{aligned} \mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] &\geq \sum_{i=1}^d \left(\frac{L_i \Delta_i \sigma_i^2}{N} \right)^{\frac{1}{4}} = \sum_{i=1}^d \left(\frac{\sigma_i^{\frac{8}{3}} L_i^{\frac{4}{3}} \Delta}{\sum_{j=1}^d \sigma_j^{\frac{2}{3}} L_j^{\frac{1}{3}} N} \right)^{\frac{1}{4}} \\ &= \left(\frac{\left(\sum_{i=1}^d \sigma_i^{\frac{2}{3}} L_i^{\frac{1}{3}} \right)^3 \Delta}{N} \right)^{\frac{1}{4}}. \end{aligned}$$

Further, by setting $L_i = (\sigma_i / \|\sigma\|_1) \|\mathbf{L}\|_1 = \frac{\sigma_i}{\|\sigma\|_1} L_\infty$, we get

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] \geq \left(\frac{\|\sigma\|_1^2 L \Delta}{N} \right)^{\frac{1}{4}}.$$

This completes the proof. $\square$

可以很容易地验证 g_t^b 满足该要求。

现在我们将 m_t 定义为在 t 步内 g_t 非零的步数，即对于 m_t 步， x_t 前进 η 步，而对于 $t - m_t$ 步，算法对 x_t 无任何操作，因为 $g_t = 0, \text{sign}(g_t) = 0$ 。我们记 $M = m_T$ 。根据定义，我们有

$$\mathbb{E}(M) = T \times \left(1 - \left(1 - \frac{4\epsilon^2}{\sigma^2 + 4\epsilon^2}\right)^B\right) \leq T \times \left(1 - 1 + \frac{4B\epsilon^2}{\sigma^2 + 4\epsilon^2}\right) = \frac{4\epsilon^2}{\sigma^2 + 4\epsilon^2}BT.$$

根据马尔可夫不等式，我们有

$$\mathbb{P}(M \geq 2\mathbb{E}[M]) \leq \frac{1}{2},$$

这意味着，以至少 1/2 的概率，我们有 $M \leq 2\mathbb{E}[M] \leq n$ 。此外，我们可以使用归纳法证明 $x_t = \eta + \eta M_{t-1}$ ，这意味着 $p'(x_t) = -2\epsilon$ 。最后，我们可以界定

$$\mathbb{E} \left[\min_t |p'(x_t)| \right] \geq \frac{1}{2} \mathbb{E} \left[\min_t |p'(x_t)| \mid M \leq n \right] = \epsilon.$$

由于上述等式对任何 $N = TB \leq L\Delta(4\epsilon^2 + \sigma^2)/128\epsilon^4$ 都成立，通过设置 $\epsilon = \sqrt{L\Delta/32N}$ ，我们得到

$$N = \frac{L\Delta}{32\epsilon^2} \leq L\Delta \frac{4\epsilon^2 + \sigma^2}{128\epsilon^4}. \quad (22)$$

类似地，我们可以设置 $\epsilon = (L\Delta\sigma^2/128N)^{1/4}$ 来满足条件。因此对于某个常数 C ，我们有

$$\mathbb{E} \left[\min_t |p'(x_t)| \right] \geq C \max \left\{ \left(\frac{L\Delta\sigma^2}{N} \right)^{\frac{1}{4}}, \sqrt{\frac{L\Delta}{N}} \right\}. \quad (23)$$

这就完成了证明。 $\square$

引理 3 给出了算法 1 对于一维函数的复杂度下界。现在我们可以将其扩展到维度 d 结果，这将得到定理 2 中的界限。

定理 2 的证明。首先，通过对每个坐标应用引理 3，我们得到

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] \geq \sum_{i=1}^d \max \left\{ \left(\frac{L_i \Delta_i \sigma_i^2}{N} \right)^{\frac{1}{4}}, \sqrt{\frac{L_i \Delta_i}{N}} \right\}.$$

通过选择 $\Delta_i = L_i \Delta / \|L\|_1$ ，我们有：

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] \geq \sum_{i=1}^d \sqrt{\frac{L_i \Delta_i}{N}} = \sum_{i=1}^d L_i \sqrt{\frac{\Delta}{\|L\|_1 N}} = \sqrt{\frac{\|L\|_1 \Delta}{N}}.$$

然后，通过选择 $\Delta_i = \frac{\sigma_i^{\frac{2}{3}} L_i^{\frac{1}{3}}}{\sum_{j=1}^d \sigma_j^{\frac{2}{3}} L_j^{\frac{1}{3}}} \Delta$ ，我们得到

$$\begin{aligned} \mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] &\geq \sum_{i=1}^d \left(\frac{L_i \Delta_i \sigma_i^2}{N} \right)^{\frac{1}{4}} = \sum_{i=1}^d \left(\frac{\sigma_i^{\frac{8}{3}} L_i^{\frac{4}{3}} \Delta}{\sum_{j=1}^d \sigma_j^{\frac{2}{3}} L_j^{\frac{1}{3}} N} \right)^{\frac{1}{4}} \\ &= \left(\frac{\left(\sum_{i=1}^d \sigma_i^{\frac{2}{3}} L_i^{\frac{1}{3}} \right)^3 \Delta}{N} \right)^{\frac{1}{4}}. \end{aligned}$$

此外，通过设置 $L_i = (\sigma_i / \|\sigma\|_1) \|\mathbf{L}\|_1 = \frac{\sigma_i}{\|\sigma\|_1} L_\infty$ ，我们得到

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] \geq \left(\frac{\|\sigma\|_1^2 L \Delta}{N} \right)^{\frac{1}{4}}.$$

这完成了证明。 $\square$

C Proof of Theorem 3

Proof. In Jiang et al. [2025a], the lower bound of SGD is established as follows.

Lemma 4. (Jiang et al. [2025a, Theorem 4.1]) Run vanilla SGD for T iterations with a batch size B , there exists a function which satisfies Assumption 1a with $f(\mathbf{x}_1) - \inf f \leq \Delta$, Assumption 4a with $\mathbf{L} = [L_1, L_2, \dots, L_d]$, and Assumption 2a with $\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \dots, \sigma_d]$, and we have:

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{d\|\mathbf{L}\|_\infty \Delta}{T}} + \left(\frac{d\Delta(\sum_{i=1}^d \sigma_i \sqrt{L_i})^2}{T} \right)^{\frac{1}{4}} \right). \quad (24)$$

Now with Lemma 1, we know that any $\mathbf{L}$ separable smooth function f is also ℓ_∞ -smooth with $L_\infty = \|\mathbf{L}\|_1$. So we can properly set L_i to maximize (24) to get the lower bound of SGD under Assumption 3a.

For the first term, let's consider a case where $L_\infty = \|\mathbf{L}\|_1$ is imbalanced and dominated by a certain coordinate, for example, we have

$$L_1 > \frac{1}{2} \|\mathbf{L}\|_1, \|\mathbf{L}\|_\infty = L_1 \geq \frac{1}{2} \|\mathbf{L}\|_1 = \frac{1}{2} L_\infty.$$

So under this case,

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{d\|\mathbf{L}\|_\infty \Delta}{T}} \right) = \Omega \left(\sqrt{\frac{dL_\infty \Delta}{T}} \right). \quad (25)$$

Then, for the second term, by setting $L_i = (\sigma_i^2 / \|\boldsymbol{\sigma}\|_2^2) L_\infty$, we get

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\left(\frac{dL_\infty \Delta \|\boldsymbol{\sigma}\|_2^2}{T} \right)^{\frac{1}{4}} \right). \quad (26)$$

Putting the two cases in (25) and (26) together and we can obtain (5). This completes the proof. $\square$

D Analysis for Muon

D.1 Upper Bound for Muon

Proof of Theorem 4. Under Assumption 3b, we have:

$$\begin{aligned} F(\mathbf{W}_{t+1}) &\leq F(\mathbf{W}_t) + \langle \nabla F(\mathbf{W}_t), \mathbf{W}_{t+1} - \mathbf{W}_t \rangle + \frac{L_*}{2} \|\mathbf{W}_{t+1} - \mathbf{W}_t\|_{\text{op}}^2 \\ &= F(\mathbf{W}_t) - \langle \nabla F(\mathbf{W}_t), \eta \text{msign}(\mathbf{G}_t) \rangle + \frac{L_*}{2} \|\eta \text{msign}(\mathbf{G}_t)\|_{\text{op}}^2 \\ &= F(\mathbf{W}_t) - \eta \langle \mathbf{G}_t, \text{msign}(\mathbf{G}_t) \rangle - \eta \langle \nabla F(\mathbf{W}_t) - \mathbf{G}_t, \text{msign}(\mathbf{G}_t) \rangle + \frac{L_* \eta^2}{2} \\ &\leq F(\mathbf{W}_t) - \eta \|\mathbf{G}_t\|_* + \eta \|\nabla F(\mathbf{W}_t) - \mathbf{G}_t\|_* \|\text{msign}(\mathbf{G}_t)\|_{\text{op}} + \frac{L_* \eta^2}{2} \\ &\leq F(\mathbf{W}_t) - \eta \|\nabla F(\mathbf{W}_t)\|_* + 2\eta \|\nabla F(\mathbf{W}_t) - \mathbf{G}_t\|_* + \frac{L_* \eta^2}{2}, \end{aligned}$$

where the second inequality is due to $|\langle A, B \rangle| \leq \|A\|_* \|B\|_{\text{op}}$. Summing over $t = 0, \dots, T-1$, we get

$$F(\mathbf{W}_T) - F(\mathbf{W}_1) \leq -\eta \sum_{t=0}^{T-1} \|\nabla F(\mathbf{W}_t)\|_* + 2\eta \sum_{t=0}^{T-1} \|\nabla F(\mathbf{W}_t) - \mathbf{G}_t\|_* + \frac{L_* \eta^2 T}{2}. \quad (27)$$

Under Assumption 1b, $F(\mathbf{W}_1) - F(\mathbf{W}_T) \leq F(\mathbf{W}_1) - \inf F \leq \Delta$, take expectation on both sides, and we get

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla F(\mathbf{W}_t)\|_*] \leq \frac{\Delta}{\eta T} + \frac{2}{T} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla F(\mathbf{W}_t) - \mathbf{G}_t\|_*] + \frac{L_* \eta}{2}. \quad (28)$$

Define $\mathbf{N}_t = \nabla F(\mathbf{W}_t) - \mathbf{G}_t$. Consider Batch size B , we denote the gradient on each sample is $\mathbf{G}_t^b$, $1 \leq b \leq B$, and the noise on each sample as $\mathbf{N}_t^b = \nabla F(\mathbf{W}_t) - \mathbf{G}_t^b$. We have the following lemma:

C 证明定理3

证明。在 姜等人 [2025一个], SGD的下界被建立如下。

引理4. (Jiang 等人 [2025a, 定理4. 1]) 运行标准 SGD, 迭代 T 次, 批大小 B , 存在一个函数满足假设1a, 其 $f(\mathbf{x}_1) - \inf f \leq \Delta$, 满足假设4a, 以及 $\mathbf{L} = [L_1, L_2, \dots, L_d]$, 满足假设2a, 并且我们有:

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{d\|\mathbf{L}\|_\infty \Delta}{T}} + \left(\frac{d\Delta(\sum_{i=1}^d \sigma_i \sqrt{L_i})^2}{T} \right)^{\frac{1}{4}} \right). \quad (24)$$

现在根据引理1, 我们知道任何 $\mathbf{L}$ 可分平滑函数 f 也是 ℓ_∞ -平滑的, 其 $L_\infty = \|\mathbf{L}\|_1$ 。因此我们可以适当设置 L_i 以最大化 (24) 来获得假设3a下 SGD 的下界。

对于第一项, 让我们考虑一个 $L_\infty = \|\mathbf{L}\|_1$ 不平衡且被某个坐标主导的情况, 例如, 我们有

$$L_1 > \frac{1}{2} \|\mathbf{L}\|_1, \|\mathbf{L}\|_\infty = L_1 \geq \frac{1}{2} \|\mathbf{L}\|_1 = \frac{1}{2} L_\infty.$$

那么在这种情况下,

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{d\|\mathbf{L}\|_\infty \Delta}{T}} \right) = \Omega \left(\sqrt{\frac{dL_\infty \Delta}{T}} \right). \quad (25)$$

接下来, 对于第二项, 通过设置 $L_i = (\sigma_i^2 / \|\boldsymbol{\sigma}\|_2^2) L_\infty$, 我们得到

$$\mathbb{E} \left[\min_t \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\left(\frac{dL_\infty \Delta \|\boldsymbol{\sigma}\|_2^2}{T} \right)^{\frac{1}{4}} \right). \quad (26)$$

将两种情况合并到 (25) 和 (26) 中, 我们可以得到 (5)。这完成了证明。 $\square$

D. Muon 分析

D.1 Muon 上限

定理 4 的证明。在假设 3b 下, 我们有:

$$\begin{aligned} F(\mathbf{W}_{t+1}) &\leq F(\mathbf{W}_t) + \langle \nabla F(\mathbf{W}_t), \mathbf{W}_{t+1} - \mathbf{W}_t \rangle + \frac{L_*}{2} \|\mathbf{W}_{t+1} - \mathbf{W}_t\|_{\text{op}}^2 \\ &= F(\mathbf{W}_t) - \langle \nabla F(\mathbf{W}_t), \eta \text{msign}(\mathbf{G}_t) \rangle + \frac{L_*}{2} \|\eta \text{msign}(\mathbf{G}_t)\|_{\text{op}}^2 \\ &= F(\mathbf{W}_t) - \eta \langle \mathbf{G}_t, \text{msign}(\mathbf{G}_t) \rangle - \eta \langle \nabla F(\mathbf{W}_t) - \mathbf{G}_t, \text{msign}(\mathbf{G}_t) \rangle + \frac{L_* \eta^2}{2} \\ &\leq F(\mathbf{W}_t) - \eta \|\mathbf{G}_t\|_* + \eta \|\nabla F(\mathbf{W}_t) - \mathbf{G}_t\|_* \|\text{msign}(\mathbf{G}_t)\|_{\text{op}} + \frac{L_* \eta^2}{2} \\ &\leq F(\mathbf{W}_t) - \eta \|\nabla F(\mathbf{W}_t)\|_* + 2\eta \|\nabla F(\mathbf{W}_t) - \mathbf{G}_t\|_* + \frac{L_* \eta^2}{2}, \end{aligned}$$

其中第二个不等式归因于 $|\langle A, B \rangle| \leq \|A\|_* \|B\|_{\text{op}}$ 。对 $t = 0, \dots, T-1$ 求和, 我们得到

$$F(\mathbf{W}_T) - F(\mathbf{W}_1) \leq -\eta \sum_{t=0}^{T-1} \|\nabla F(\mathbf{W}_t)\|_* + 2\eta \sum_{t=0}^{T-1} \|\nabla F(\mathbf{W}_t) - \mathbf{G}_t\|_* + \frac{L_* \eta^2 T}{2}. \quad (27)$$

在假设 1b下, $F(\mathbf{W}_1) - F(\mathbf{W}_T) \leq F(\mathbf{W}_1) - \inf F \leq \Delta$, 对两边取期望, 我们得到

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla F(\mathbf{W}_t)\|_*] \leq \frac{\Delta}{\eta T} + \frac{2}{T} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla F(\mathbf{W}_t) - \mathbf{G}_t\|_*] + \frac{L_* \eta}{2}. \quad (28)$$

定义 $\mathbf{N}_t = \nabla F(\mathbf{W}_t) - \mathbf{G}_t$ 。考虑批大小 B , 我们记每个样本上的梯度为 $\mathbf{G}_t^b, 1 \leq b \leq B$, 噪声为 $\mathbf{N}_t^b = \nabla F(\mathbf{W}_t) - \mathbf{G}_t^b$ 。我们有以下引理:

Lemma 5. (An et al. [2025, Lemma 9]) Under Assumption 2b, with batch size B , we have

$$\mathbb{E} [\mathbf{N}_t \mathbf{N}_t^\top] \preceq \frac{\boldsymbol{\Sigma}^2}{B}. \quad (29)$$

Under the fact that the map $X \mapsto \text{Tr} \left(X^{\frac{1}{2}} \right)$ is concave on the positive semi-definite cone, we can apply Jensen's inequality to get

$$\mathbb{E} [\|\mathbf{N}_t\|_*] = \mathbb{E} \left[\text{Tr} \left(\sqrt{\mathbf{N}_t \mathbf{N}_t^\top} \right) \right] \leq \text{Tr} \left(\sqrt{\mathbb{E} [\mathbf{N}_t \mathbf{N}_t^\top]} \right) \leq \frac{\|\boldsymbol{\Sigma}\|_*}{\sqrt{B}}. \quad (30)$$

Plugging (30) back to (28) and set $\eta = \sqrt{2\Delta/L_*T}$, we get

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla F(\mathbf{W}_t)\|_*] \leq \sqrt{\frac{2\Delta L_*}{T}} + 2 \frac{\|\boldsymbol{\Sigma}\|_*}{\sqrt{B}}. \quad (31)$$

If $T \leq \Delta L_*/\|\boldsymbol{\Sigma}\|_*^2$, By setting $B = 1$, we have

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla F(\mathbf{W}_t)\|_* \right] \leq \sqrt{\frac{2\Delta L_*}{T}} + 2\|\boldsymbol{\Sigma}\|_* \leq (2 + \sqrt{2}) \sqrt{\frac{\Delta L_*}{N}}$$

Otherwise, we set $B = \|\boldsymbol{\Sigma}\|_*^2/\Delta L_*T$, which implies

$$\sqrt{\frac{2\Delta L_*}{T}} = \left(\frac{4\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{BT} \right)^{\frac{1}{4}}. \quad (32)$$

Bringing (32) back to (31)

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla F(\mathbf{W}_t)\|_* \right] &\leq 2 \left(\frac{\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{BT} \right)^{\frac{1}{4}} + \left(\frac{4\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{BT} \right)^{\frac{1}{4}} \\ &= \mathcal{O} \left(\left(\frac{\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{BT} \right)^{\frac{1}{4}} \right). \end{aligned}$$

Putting the above two cases together, we have

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla F(\mathbf{W}_t)\|_* \right] = \mathcal{O} \left(\sqrt{\frac{\Delta L_*}{N}} + \left(\frac{\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{N} \right)^{\frac{1}{4}} \right).$$

□

D.2 Lower Bound for Muon

Assume the target weight matrix is $\mathbf{W} \in \mathbb{R}^{m \times n}$. Without loss of generality, we assume $m \leq n$. We introduce a projection matrix $\mathbf{P} = [\mathbf{I}_m, \mathbf{0}_{m \times (n-m)}] \in \mathbb{R}^{m \times n}$ which pads an $m \times m$ matrix with zeros to match the $m \times n$ dimension. Let $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times m}$ be the target matrix noise covariance, which can be diagonalized via an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{m \times m}$ as $\boldsymbol{\Sigma} = \mathbf{Q}^\top \text{diag}(\boldsymbol{\sigma}) \mathbf{Q}$.

We construct our worst-case matrix objective function by extracting the diagonal elements of the transformed matrix $\mathbf{QWP}^\top \in \mathbb{R}^{m \times m}$:

$$F(\mathbf{W}) = \sum_{i=1}^m f_i((\mathbf{QWP}^\top)_{ii}), \quad (33)$$

where $f(\mathbf{x}) = \sum_{i=1}^m f_i(\mathbf{x}_i)$ is the hard 1D vector instance constructed in Theorem 2.

First, we provide two lemmas to establish the strict geometric equivalence between the vector domain and the matrix domain.

Lemma 6. Assume that the separable vector function $f(\mathbf{x}) : \mathbb{R}^m \rightarrow \mathbb{R}$ satisfies Assumption 2a with bounded variance σ^2 . Then the constructed matrix function $F(\mathbf{W}) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ satisfies Assumption 2b with matrix noise covariance $\boldsymbol{\Sigma}^2 = \mathbf{Q}^\top \text{diag}(\sigma^2) \mathbf{Q}$.

引理 5. (An et al. [2025, 引理 9]) 在假设 2b 下, 当批大小为 B , 我们有

$$\mathbb{E} [\mathbf{N}_t \mathbf{N}_t^\top] \preceq \frac{\boldsymbol{\Sigma}^2}{B}. \quad (29)$$

由于映射 $X \mapsto \text{Tr} \left(X^{\frac{1}{2}} \right)$ 在正半定锥上是凹的, 我们可以应用 Jensen 不等式得到

$$\mathbb{E} [\|\mathbf{N}_t\|_*] = \mathbb{E} \left[\text{Tr} \left(\sqrt{\mathbf{N}_t \mathbf{N}_t^\top} \right) \right] \leq \text{Tr} \left(\sqrt{\mathbb{E} [\mathbf{N}_t \mathbf{N}_t^\top]} \right) \leq \frac{\|\boldsymbol{\Sigma}\|_*}{\sqrt{B}}. \quad (30)$$

将 (30) 代入 (28) 并设 $\eta = \sqrt{2\Delta/L_*T}$, 我们得到

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla F(\mathbf{W}_t)\|_*] \leq \sqrt{\frac{2\Delta L_*}{T}} + 2 \frac{\|\boldsymbol{\Sigma}\|_*}{\sqrt{B}}. \quad (31)$$

如果 $T \leq \Delta L_*/\|\boldsymbol{\Sigma}\|_*^2$, 通过设定 $B = 1$, 我们有

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla F(\mathbf{W}_t)\|_* \right] \leq \sqrt{\frac{2\Delta L_*}{T}} + 2\|\boldsymbol{\Sigma}\|_* \leq (2 + \sqrt{2}) \sqrt{\frac{\Delta L_*}{N}}$$

否则, 我们设 $B = \|\boldsymbol{\Sigma}\|_*^2/\Delta L_*T$, 这意味着

$$\sqrt{\frac{2\Delta L_*}{T}} = \left(\frac{4\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{BT} \right)^{\frac{1}{4}}. \quad (32)$$

将 (32) 代入 (31)

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla F(\mathbf{W}_t)\|_* \right] &\leq 2 \left(\frac{\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{BT} \right)^{\frac{1}{4}} + \left(\frac{4\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{BT} \right)^{\frac{1}{4}} \\ &= \mathcal{O} \left(\left(\frac{\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{BT} \right)^{\frac{1}{4}} \right). \end{aligned}$$

将上述两种情况结合起来, 我们有

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla F(\mathbf{W}_t)\|_* \right] = \mathcal{O} \left(\sqrt{\frac{\Delta L_*}{N}} + \left(\frac{\|\boldsymbol{\Sigma}\|_*^2 \Delta L_*}{N} \right)^{\frac{1}{4}} \right).$$

□

D.2 Muon 下界

假设目标权重矩阵为 $\mathbf{W} \in \mathbb{R}^{m \times n}$. 不失一般性, 我们假设 $m \leq n$. 我们引入一个投影矩阵 $\mathbf{P} \in \mathbb{R}^{m \times n}$ 它通过用零填充矩阵来匹配 $\mathbf{W}$ 的维度. 设 $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times m}$ 为目标矩阵噪声协方差, 它可以通过正交矩阵 $\mathbf{Q} \in \mathbb{R}^{m \times m}$ 对角化, 即 $\boldsymbol{\Sigma} = \mathbf{Q}^\top \text{diag}(\boldsymbol{\sigma}) \mathbf{Q}$. 我们通过提取变换矩阵 $\mathbf{QWP}^\top \in \mathbb{R}^{m \times m}$ 的对角元素来构建我们的最坏情况矩阵目标函数:

$$F(\mathbf{W}) = \sum_{i=1}^m f_i((\mathbf{QWP}^\top)_{ii}), \quad (33)$$

其中 $f(\mathbf{x}) = \sum_{i=1}^m f_i(\mathbf{x}_i)$ 是在定理 2 中构造的硬 1D 向量实例.

首先, 我们提供两个引理来建立向量域和矩阵域之间的严格几何等价性.

引理 6. 假设可分离向量函数 $f(\mathbf{x}) : \mathbb{R}^m \rightarrow \mathbb{R}$ 满足假设 2a 且具有有界方差 σ^2 . 那么构造的矩阵函数 $F(\mathbf{W}) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ 满足假设 2b 且具有矩阵噪声协方差 $\boldsymbol{\Sigma}^2 = \mathbf{Q}^\top \text{diag}(\sigma^2) \mathbf{Q}$.

Proof. By the chain rule, the exact gradient of $F(\mathbf{W})$ is given by:

$$\nabla F(\mathbf{W}) = \mathbf{Q}^\top \text{diag}(\nabla f(\mathbf{x}))\mathbf{P}, \quad (34)$$

where $\mathbf{x} \in \mathbb{R}^m$ with $\mathbf{x}_i = (\mathbf{Q}\mathbf{W}\mathbf{P}^\top)_{ii}$. Let $\mathbf{g} \in \mathbb{R}^m$ be the stochastic gradient oracle for $f(\mathbf{x})$. We construct the corresponding stochastic matrix gradient $\mathbf{G}$ for $F(\mathbf{W})$ as:

$$\mathbf{G} = \mathbf{Q}^\top \text{diag}(\mathbf{g})\mathbf{P}. \quad (35)$$

The estimation error matrix is $\mathbf{G} - \nabla F(\mathbf{W}) = \mathbf{Q}^\top \text{diag}(\mathbf{g} - \nabla f(\mathbf{x}))\mathbf{P}$. Computing the uncentered covariance, we have:

$$\begin{aligned} \mathbb{E}[(\mathbf{G} - \nabla F(\mathbf{W}))(\mathbf{G} - \nabla F(\mathbf{W}))^\top] &= \mathbb{E}[\mathbf{Q}^\top \text{diag}(\mathbf{g} - \nabla f(\mathbf{x}))\mathbf{P}\mathbf{P}^\top \text{diag}(\mathbf{g} - \nabla f(\mathbf{x}))\mathbf{Q}] \\ &= \mathbf{Q}^\top \mathbb{E}[\text{diag}((g_i - \nabla f_i(\mathbf{x}_i))^2)]\mathbf{Q}. \end{aligned}$$

Here we used the property that $\mathbf{P}\mathbf{P}^\top = \mathbf{I}_m$. Under Assumption 2a, we have $\mathbb{E}[(g_i - \nabla f_i(\mathbf{x}_i))^2] \leq \sigma_i^2$, which implies:

$$\mathbb{E}[(\mathbf{G} - \nabla F(\mathbf{W}))(\mathbf{G} - \nabla F(\mathbf{W}))^\top] \preceq \mathbf{Q}^\top \text{diag}(\sigma^2)\mathbf{Q} = \Sigma^2.$$

This completes the proof. $\square$

Lemma 7. Assume that $f(\mathbf{x}) : \mathbb{R}^m \rightarrow \mathbb{R}$ satisfies Assumption 3a with L_∞ -smoothness parameter L . Then the constructed matrix function $F(\mathbf{W}) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ satisfies Assumption 3b with spectral norm smoothness $L_* = L$.

Proof. Let $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{m \times n}$ be two arbitrary matrices, and define $\Delta = \mathbf{Y} - \mathbf{X}$. We map these matrices to vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^m$ by defining $\mathbf{x}_i = (\mathbf{Q}\mathbf{X}\mathbf{P}^\top)_{ii}$ and $\mathbf{y}_i = (\mathbf{Q}\mathbf{Y}\mathbf{P}^\top)_{ii}$. Let their difference be $\delta = \mathbf{y} - \mathbf{x}$, where $\delta_i = (\mathbf{Q}\Delta\mathbf{P}^\top)_{ii}$. By definition, $F(\mathbf{X}) = f(\mathbf{x})$ and $F(\mathbf{Y}) = f(\mathbf{y})$.

First, we bridge the Frobenius inner product of the matrices with the Euclidean inner product of the vectors. Let $\mathbf{S} = \text{diag}(\nabla f(\mathbf{x}))$. Using (34) and the cyclic property of the trace operator, we have:

$$\begin{aligned} \langle \nabla F(\mathbf{X}), \Delta \rangle_F &= \text{tr}((\mathbf{Q}^\top \mathbf{S} \mathbf{P})^\top \Delta) \\ &= \text{tr}(\mathbf{P}^\top \mathbf{S}^\top \mathbf{Q} \Delta) \\ &= \text{tr}(\mathbf{S}^\top (\mathbf{Q} \Delta \mathbf{P}^\top)). \end{aligned}$$

Let $\mathbf{M} = \mathbf{Q} \Delta \mathbf{P}^\top \in \mathbb{R}^{m \times m}$. Since $\mathbf{S}$ is diagonal, the trace $\text{tr}(\mathbf{S}^\top \mathbf{M})$ perfectly reduces to the element-wise sum of their main diagonals:

$$\text{tr}(\mathbf{S}^\top \mathbf{M}) = \sum_{i=1}^m \mathbf{S}_{ii} \mathbf{M}_{ii} = \sum_{i=1}^m \nabla f_i(\mathbf{x}_i) \delta_i = \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle.$$

Therefore, the Taylor expansion residuals in both domains are strictly identical:

$$F(\mathbf{Y}) - (F(\mathbf{X}) + \langle \nabla F(\mathbf{X}), \mathbf{Y} - \mathbf{X} \rangle_F) = f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle). \quad (36)$$

Taking the absolute value and applying the L_∞ -smoothness of $f(\mathbf{x})$, we obtain:

$$|F(\mathbf{Y}) - F(\mathbf{X}) - \langle \nabla F(\mathbf{X}), \mathbf{Y} - \mathbf{X} \rangle_F| \leq \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|_\infty^2. \quad (37)$$

To bound the vector ℓ_∞ -norm using the matrix spectral norm, we examine the maximum component of δ . For any $i \in \{1, \dots, m\}$:

$$|\mathbf{y}_i - \mathbf{x}_i| = |(\mathbf{Q} \Delta \mathbf{P}^\top)_{ii}| = |\mathbf{e}_i^\top \mathbf{Q} \Delta \mathbf{P}^\top \mathbf{e}_i| = |\mathbf{q}_i^\top \Delta \tilde{\mathbf{e}}_i|,$$

where $\mathbf{e}_i \in \mathbb{R}^m$ is the standard basis vector, $\mathbf{q}_i = \mathbf{Q}^\top \mathbf{e}_i \in \mathbb{R}^m$, and $\tilde{\mathbf{e}}_i = \mathbf{P}^\top \mathbf{e}_i \in \mathbb{R}^n$. Since $\mathbf{Q}$ is orthogonal and $\mathbf{P}$ is a standard projection, both vectors preserve their unit Euclidean lengths: $\|\mathbf{q}_i\|_2 = 1$ and $\|\tilde{\mathbf{e}}_i\|_2 = 1$. By the definition of the induced matrix 2-norm (spectral norm), we have:

$$|\mathbf{q}_i^\top \Delta \tilde{\mathbf{e}}_i| \leq \|\mathbf{q}_i\|_2 \|\Delta\|_{\text{op}} \|\tilde{\mathbf{e}}_i\|_2 = \|\mathbf{Y} - \mathbf{X}\|_{\text{op}}.$$

Taking the maximum over all i yields $\|\mathbf{y} - \mathbf{x}\|_\infty \leq \|\mathbf{Y} - \mathbf{X}\|_{\text{op}}$. Substituting this into (37) establishes that $F(\mathbf{W})$ is L_* -spectral norm smooth with $L_* = L$. $\square$

证明。通过链式法则， $F(\mathbf{W})$ 的精确梯度为：

$$\nabla F(\mathbf{W}) = \mathbf{Q}^\top \text{diag}(\nabla f(\mathbf{x}))\mathbf{P}, \quad (34)$$

在 $\mathbf{x} \in \mathbb{R}^m$ 处，其中 $\mathbf{x}_i = (\mathbf{Q}\mathbf{W}\mathbf{P}^\top)_{ii}$ 。令 $\mathbf{g} \in \mathbb{R}^m$ 为 $f(\mathbf{x})$ 的随机梯度预言机。我们为 $F(\mathbf{W})$ 构造相应的随机矩阵梯度 $\mathbf{G}$ ，如下：

$$\mathbf{G} = \mathbf{Q}^\top \text{diag}(\mathbf{g})\mathbf{P}. \quad (35)$$

估计误差矩阵为 $\mathbf{G} - \nabla F(\mathbf{W}) = \mathbf{Q}^\top \text{diag}(\mathbf{g} - \nabla f(\mathbf{x}))\mathbf{P}$ 。计算非中心协方差，我们有：

$$\begin{aligned} \mathbb{E}[(\mathbf{G} - \nabla F(\mathbf{W}))(\mathbf{G} - \nabla F(\mathbf{W}))^\top] &= \mathbb{E}[\mathbf{Q}^\top \text{diag}(\mathbf{g} - \nabla f(\mathbf{x}))\mathbf{P}\mathbf{P}^\top \text{diag}(\mathbf{g} - \nabla f(\mathbf{x}))\mathbf{Q}] \\ &= \mathbf{Q}^\top \mathbb{E}[\text{diag}((g_i - \nabla f_i(\mathbf{x}_i))^2)]\mathbf{Q}. \end{aligned}$$

这里我们使用了 $\mathbf{P}\mathbf{P}^\top = \mathbf{I}_m$ 的性质。在假设 2a 下，我们有 $\mathbb{E}[(g_i - \nabla f_i(\mathbf{x}_i))^2] \leq \sigma_i^2$ ，这意味着：

$$\mathbb{E}[(\mathbf{G} - \nabla F(\mathbf{W}))(\mathbf{G} - \nabla F(\mathbf{W}))^\top] \preceq \mathbf{Q}^\top \text{diag}(\sigma^2)\mathbf{Q} = \Sigma^2.$$

这完成了证明。 $\square$

引理 7。 假设 $f(\mathbf{x}) : \mathbb{R}^m \rightarrow \mathbb{R}$ 满足假设 3a，且 L_∞ -范数平滑性参数 L 。那么构造的矩阵函数 $F(\mathbf{W}) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ 满足假设 3b，且具有谱范数平滑性 $L_* = L$ 。

证明。令 $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{m \times n}$ 为两个任意矩阵，并定义 $\Delta = \mathbf{Y} - \mathbf{X}$ 。我们将这些矩阵映射为向量 $\mathbf{x}, \mathbf{y} \in \mathbb{R}^m$ ，通过定义 $\mathbf{x}_i = (\mathbf{Q}\mathbf{X}\mathbf{P}^\top)_{ii}$ 和 $\mathbf{y}_i = (\mathbf{Q}\mathbf{Y}\mathbf{P}^\top)_{ii}$ 。令它们的差为 $\delta = \mathbf{y} - \mathbf{x}$ ，其中 $\delta_i = (\mathbf{Q}\Delta\mathbf{P}^\top)_{ii}$ 。根据定义， $F(\mathbf{X}) = f(\mathbf{x})$ 且 $F(\mathbf{Y}) = f(\mathbf{y})$ 。

首先，我们将矩阵的Frobenius内积与向量的欧几里得内积联系起来。令 $\mathbf{S} = \text{diag}(\nabla f(\mathbf{x}))$ 。利用 (34)和迹算子的循环性质，我们得到：

$$\begin{aligned} \langle \nabla F(\mathbf{X}), \Delta \rangle_F &= \text{tr}((\mathbf{Q}^\top \mathbf{S} \mathbf{P})^\top \Delta) \\ &= \text{tr}(\mathbf{P}^\top \mathbf{S}^\top \mathbf{Q} \Delta) \\ &= \text{tr}(\mathbf{S}^\top (\mathbf{Q} \Delta \mathbf{P}^\top)). \end{aligned}$$

令 $\mathbf{M} = \mathbf{Q} \Delta \mathbf{P}^\top \in \mathbb{R}^{m \times m}$ 。由于 $\mathbf{S}$ 是对角矩阵， $\text{tr}(\mathbf{S}^\top \mathbf{M})$ 完美地简化为它们主对角线元素的和：

$$\text{tr}(\mathbf{S}^\top \mathbf{M}) = \sum_{i=1}^m \mathbf{S}_{ii} \mathbf{M}_{ii} = \sum_{i=1}^m \nabla f_i(\mathbf{x}_i) \delta_i = \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle.$$

因此，两个域中的泰勒展开残差严格相同：

$$F(\mathbf{Y}) - (F(\mathbf{X}) + \langle \nabla F(\mathbf{X}), \mathbf{Y} - \mathbf{X} \rangle_F) = f(\mathbf{y}) - (f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle). \quad (36)$$

取绝对值并应用 L_∞ 次平滑性 $f(\mathbf{x})$ ，我们得到：

$$|F(\mathbf{Y}) - F(\mathbf{X}) - \langle \nabla F(\mathbf{X}), \mathbf{Y} - \mathbf{X} \rangle_F| \leq \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|_\infty^2. \quad (37)$$

为使用矩阵谱范数界定向量 ℓ_∞ -范数，我们考察 δ 的最大分量。对于任何 $i \in \{1, \dots, m\}$ ：

$$|\mathbf{y}_i - \mathbf{x}_i| = |(\mathbf{Q} \Delta \mathbf{P}^\top)_{ii}| = |\mathbf{e}_i^\top \mathbf{Q} \Delta \mathbf{P}^\top \mathbf{e}_i| = |\mathbf{q}_i^\top \Delta \tilde{\mathbf{e}}_i|,$$

其中 $\mathbf{e}_i \in \mathbb{R}^m$ 是标准基向量， $\mathbf{q}_i = \mathbf{Q}^\top \mathbf{e}_i \in \mathbb{R}^m$ ，且 $\tilde{\mathbf{e}}_i = \mathbf{P}^\top \mathbf{e}_i \in \mathbb{R}^n$ 。由于 $\mathbf{Q}$ 是正交矩阵且 $\mathbf{P}$ 是标准投影，这两个向量都保持其单位欧几里得长度： $\|\mathbf{q}_i\|_2 = 1$ 和 $\|\tilde{\mathbf{e}}_i\|_2 = 1$ 。根据诱导矩阵2-范数（谱范数）的定义，我们有：

$$|\mathbf{q}_i^\top \Delta \tilde{\mathbf{e}}_i| \leq \|\mathbf{q}_i\|_2 \|\Delta\|_{\text{op}} \|\tilde{\mathbf{e}}_i\|_2 = \|\mathbf{Y} - \mathbf{X}\|_{\text{op}}.$$

在所有 i 中取最大值 得到 $\|\mathbf{y} - \mathbf{x}\|_\infty \leq \|\mathbf{Y} - \mathbf{X}\|_{\text{op}}$ 。将此代入 (37) 即可证明 $F(\mathbf{W})$ 是 L_* -谱范数光滑的，且 $L_* = L$ 。 $\square$

Proof of Theorem 5. We now demonstrate that under this construction, the optimization trajectory of Muon in the matrix space exactly replicates that of SignSGD in the vector space.

Recall the constructed stochastic matrix gradient is $\mathbf{G}_t = \mathbf{Q}^\top \text{diag}(\mathbf{g}_t) \mathbf{P}$. The Muon optimizer updates the weight matrix $\mathbf{W}$ by moving in the direction of the orthogonalized gradient, computed via the thin SVD: $\mathbf{G}_t = \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top$, where the update direction is $\text{msign}(\mathbf{G}_t) = \mathbf{U} \mathbf{V}^\top$.

We can explicitly write the SVD of our constructed $\mathbf{G}_t$. Let $\mathbf{S} = \text{diag}(\text{sign}(\mathbf{g}_t))$ and $\mathbf{\Lambda} = \text{diag}(|\mathbf{g}_t|)$. We have:

$$\mathbf{G}_t = \mathbf{Q}^\top \mathbf{S} \mathbf{\Lambda} \mathbf{P} = (\mathbf{Q}^\top \text{diag}(\text{sign}(\mathbf{g}_t))) \text{diag}(|\mathbf{g}_t|) \mathbf{P}.$$

Notice that $\mathbf{Q}^\top \text{diag}(\text{sign}(\mathbf{g}_t))$ is an $m \times m$ matrix with orthonormal columns (since both $\mathbf{Q}^\top$ and $\text{diag}(\pm 1)$ are orthogonal), $\text{diag}(|\mathbf{g}_t|)$ is an $m \times m$ non-negative diagonal matrix, and $\mathbf{P}$ is an $m \times n$ matrix with orthonormal rows. This strictly matches the definition of the thin SVD. Thus, the orthogonal factor is uniquely identified as:

$$\text{msign}(\mathbf{G}_t) = \mathbf{U} \mathbf{V}^\top = \mathbf{Q}^\top \text{diag}(\text{sign}(\mathbf{g}_t)) \mathbf{P}.$$

The Muon update rule $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \text{msign}(\mathbf{G}_t)$ then translates to:

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \mathbf{Q}^\top \text{diag}(\text{sign}(\mathbf{g}_t)) \mathbf{P}.$$

Left-multiplying by $\mathbf{Q}$ and right-multiplying by $\mathbf{P}^\top$, we obtain:

$$\mathbf{Q} \mathbf{W}_{t+1} \mathbf{P}^\top = \mathbf{Q} \mathbf{W}_t \mathbf{P}^\top - \eta \text{diag}(\text{sign}(\mathbf{g}_t)).$$

By isolating the diagonal entries $\mathbf{x}_{t,i} = (\mathbf{Q} \mathbf{W}_t \mathbf{P}^\top)_{ii}$, we recover the exact SignSGD coordinate-wise update:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \text{sign}(\mathbf{g}_t).$$

This establishes a strict dynamic isomorphism: the sequence of the extracted diagonal entries perfectly replicates the trajectory of SignSGD running on the vector function $f(\mathbf{x})$.

By Lemma 6 and Lemma 7, $F(\mathbf{W})$ satisfies Assumption 2b with noise covariance $\mathbf{\Sigma} = \mathbf{Q}^\top \text{diag}(\boldsymbol{\sigma}) \mathbf{Q}$ and Assumption 3b with $L_* = L_\infty$. Furthermore, the trace norm (nuclear norm) of the noise covariance matrix strictly aligns: $\|\mathbf{\Sigma}\|_* = \text{tr}(\mathbf{\Sigma}) = \|\boldsymbol{\sigma}\|_1$.

Finally, observe the true gradient $\nabla F(\mathbf{W}_t) = \mathbf{Q}^\top \text{diag}(\nabla f(\mathbf{x}_t)) \mathbf{P}$. Because orthogonal transformations preserve singular values, the singular values of $\nabla F(\mathbf{W}_t)$ are exactly the absolute values of the components of $\nabla f(\mathbf{x}_t)$. Thus, the nuclear norm of the matrix gradient equals the ℓ_1 -norm of the vector gradient:

$$\|\nabla F(\mathbf{W}_t)\|_* = \|\nabla f(\mathbf{x}_t)\|_1.$$

Because the optimization trajectories are identical and all bounding metrics perfectly correspond, we can directly invoke the vector lower bound from Theorem 2 to establish the matrix lower bound for Muon:

$$\mathbb{E} \left[\min_t \|\nabla F(\mathbf{W}_t)\|_* \right] = \Omega \left(\sqrt{\frac{L_* \Delta}{N}} + \left(\frac{\|\mathbf{\Sigma}\|_*^2 L_* \Delta}{N} \right)^{\frac{1}{4}} \right).$$

This completes the proof. $\square$

E Experimental Details

In this section, we will present the omitted details for experiments.

E.1 Experimental Details for LLMs

Here, we present the omitted details for the experiments in Section 4.2.2 and Figure 2a. We conduct all experiments using PyTorch and Distributed Data Parallel (DDP) across four NVIDIA Pro 6000 GPUs (96GB VRAM each). Our results are based on the codebase provided by Semenov et al. [2025], which can be found at <https://github.com/epfml/llm-optimizer-benchmark/tree/main/scripts>. The GPT2-small model is trained for 10k steps with a global batch size of 512 sequences, and we use a standard sequence length of 512, thus totaling approximately $1 \times$ the Chinchilla-optimal token count as suggested by Hoffmann et al. [2022]. The best learning rates of SignSGD and SGD are found via grid search in $\{1e-2, 1e-3, 1e-4\}$. We employ a linear warm-up period of 10% total iterations at the start of pretraining.

定理5的证明。我们现在证明，在此构造下，Muon在矩阵空间中的优化轨迹精确地复制了SignSGD在向量空间中的优化轨迹。

回想一下构建的随机矩阵梯度是 $\mathbf{G}_t = \mathbf{Q}^\top \text{diag}(\mathbf{g}_t) \mathbf{P}$ 。Muon优化器通过正交化梯度的方向更新权重矩阵 $\mathbf{W}$ ，该方向通过薄SVD计算得到： $\mathbf{G}_t = \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top$ ，其中更新方向是 $\text{msign}(\mathbf{G}_t) = \mathbf{U} \mathbf{V}^\top$ 。

We 可以显式写出我们构建的 $\mathbf{G}_t$ 的SVD。令 $\mathbf{S} = \text{diag}(\text{sign}(\mathbf{g}_t))$ 和 $\mathbf{\Lambda} = \text{diag}(|\mathbf{g}_t|)$ 。We 有：

$$\mathbf{G}_t = \mathbf{Q}^\top \mathbf{S} \mathbf{\Lambda} \mathbf{P} = (\mathbf{Q}^\top \text{diag}(\text{sign}(\mathbf{g}_t))) \text{diag}(|\mathbf{g}_t|) \mathbf{P}.$$

注意， $\mathbf{Q}^\top \text{diag}(\text{sign}(\mathbf{g}_t))$ 是一个 $m \times m$ 列正交的矩阵（因为 $\mathbf{Q}^\top$ 和 $\text{diag}(\pm 1)$ 都是正交的）， $\text{diag}(|\mathbf{g}_t|)$ 是一个 $m \times m$ 非负对角矩阵，而 $\mathbf{P}$ 是一个 $m \times n$ 行正交的矩阵。这严格符合薄SVD的定义。因此，正交因子被唯一确定为：

$$\text{msign}(\mathbf{G}_t) = \mathbf{U} \mathbf{V}^\top = \mathbf{Q}^\top \text{diag}(\text{sign}(\mathbf{g}_t)) \mathbf{P}.$$

Muon更新规则 $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \text{msign}(\mathbf{G}_t)$ 翻译为：

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \mathbf{Q}^\top \text{diag}(\text{sign}(\mathbf{g}_t)) \mathbf{P}.$$

用 $\mathbf{Q}$ 从左侧相乘并用 $\mathbf{P}^\top$ 从右侧相乘，我们得到：

$$\mathbf{Q} \mathbf{W}_{t+1} \mathbf{P}^\top = \mathbf{Q} \mathbf{W}_t \mathbf{P}^\top - \eta \text{diag}(\text{sign}(\mathbf{g}_t)).$$

通过隔离对角线元素 $\mathbf{x}_{t,i} = (\mathbf{Q} \mathbf{W}_t \mathbf{P}^\top)_{ii}$ ，我们恢复了 SignSGD 的精确坐标更新：

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \text{sign}(\mathbf{g}_t).$$

这建立了一个严格的动态同构：提取的对角线元素的序列完美地复制了 SignSGD 在向量函数 $f(\mathbf{x})$ 上运行的轨迹。

根据引理 6 和引理 7, $F(\mathbf{W})$ 满足假设 2b，噪声协方差为 $\mathbf{\Sigma} = \mathbf{Q}^\top \text{diag}(\boldsymbol{\sigma}) \mathbf{Q}$ ，并满足假设 3b，协方差为 $L_* = L_\infty$ 。此外，噪声协方差矩阵的迹范数（核范数）严格对齐： $\|\mathbf{\Sigma}\|_* = \text{tr}(\mathbf{\Sigma}) = \|\boldsymbol{\sigma}\|_1$ 。

最后，观察真实梯度 $\nabla F(\mathbf{W}_t) = \mathbf{Q}^\top \text{diag}(\nabla f(\mathbf{x}_t)) \mathbf{P}$ 。由于正交变换保持奇异值， $\nabla F(\mathbf{W}_t)$ 的奇异值正好是 $\nabla f(\mathbf{x}_t)$ 分量的绝对值。因此，矩阵梯度的核范数等于向量梯度的 ℓ_1 -范数：

$$\|\nabla F(\mathbf{W}_t)\|_* = \|\nabla f(\mathbf{x}_t)\|_1.$$

因为优化轨迹相同，所有边界指标完美对应，我们可以直接调用定理2中的向量下界来建立 Muon的矩阵下界：

$$\mathbb{E} \left[\min_t \|\nabla F(\mathbf{W}_t)\|_* \right] = \Omega \left(\sqrt{\frac{L_* \Delta}{N}} + \left(\frac{\|\mathbf{\Sigma}\|_*^2 L_* \Delta}{N} \right)^{\frac{1}{4}} \right).$$

这完成了证明。 $\square$

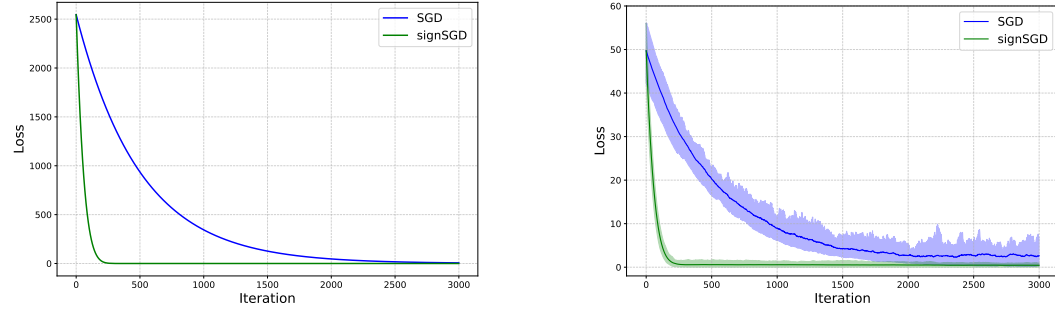
E 实验细节

在本节中，我们将介绍实验中省略的细节。

E.1 LLMs的实验细节

在此，我们呈现了第 4.2.2 节和图 2a中省略的实验细节。我们使用 PyTorch 和分布式数据并行（DDP）在四块 NVIDIA Pro 6000 GPU（每块 96GB 显存）上运行所有实验。我们的结果基于 Semenov 等人 [2025] 提供的代码库，可在

<https://github.com/epfml/llm-optimizer-benchmark/tree/main/scripts> 找到。GPT2-small 模型以 512 个序列的全局批大小训练 10k 步，我们使用标准序列长度为 512，因此总约等于 $1 \times$ Hoffmann 等人 建议的 Chinchilla-optimal token 数量。SignSGD 和 SGD 的最佳学习率通过网格搜索在 $\{1e-2, 1e-3, 1e-4\}$ 中找到。我们在预训练开始时采用 10% 总迭代数的线性预热期。



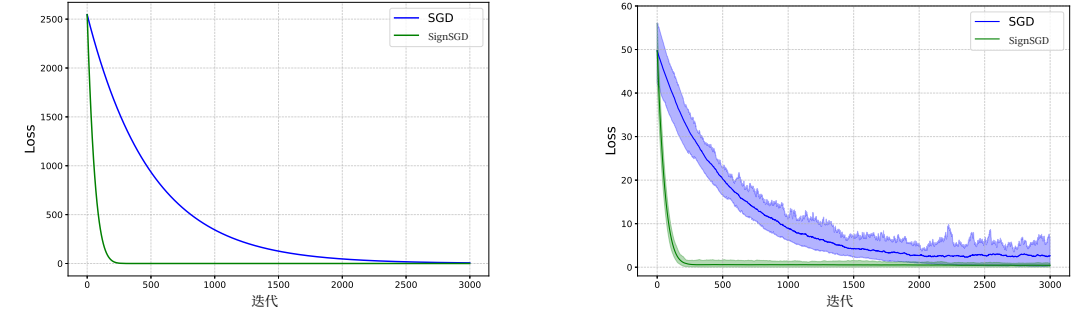
(a) Deterministic case with anisotropic curvature.

(b) Stochastic example with sparse noise.

Figure 3: Empirical toy problems demonstrating settings where SignSGD converges provably faster than SGD. Figure 3a illustrates the deterministic convergence on an imbalanced quadratic objective $f(x) = \sum_{i=1}^d L_i x_i^2 / 2$ for $x \in \mathbb{R}^{5000}$, where the curvature is dominated by a single dimension ($L_1 = 1000, L_{i>1} = 1$). This highly skewed ℓ_∞ -smooth geometry severely bottlenecks the learning rate of SGD, whereas SignSGD maintains steady convergence. Figure 3b demonstrates the training loss curve on a simple isotropic quadratic objective $f(x) = \|x\|^2 / 2$ for $x \in \mathbb{R}^{100}$, where we inject Gaussian noise $\mathcal{N}(0, 100^2)$ exclusively into the first gradient component to simulate extreme noise sparsity.

E.2 Experimental Details for CNNs

Here, we present the omitted details for the experiments on Figure 2b. We followed the codebase in Bernstein et al. [2018], which can be found at <https://github.com/jxbz/signSGD>. To ensure a fully standard and exact mathematical comparison with our LLM framework, we ported their precise architectural definitions to modern PyTorch and implemented a fully distributed variance-tracking system using PyTorch’s Distributed Data Parallel (DDP). Then, we conduct all experiments across four NVIDIA Pro 6000 GPUs (96GB VRAM each).



(a) 各向异性曲率的确定性情况。

(b) 具有稀疏噪声的随机示例。

图 3: 展示 SignSGD 在某些设置下可证明比 SGD 更快收敛的实验性玩具问题。图 3a 展示了在 $f(x) = \sum_{i=1}^d L_i x_i^2 / 2$ 上的确定性收敛，该问题是不平衡二次目标 $x \in \mathbb{R}^{5000}$ ，其中曲率由单个维度 ($L_1 = 1000, L_{i>1} = 1$) 主导。这种高度偏斜的 ℓ_∞ -光滑几何形状严重限制了 SGD 的学习率，而 SignSGD 仍保持稳定收敛。图 3b 展示了在简单各向同性二次目标 $f(x) = \|x\|^2 / 2$ 上的训练损失曲线，其中 $x \in \mathbb{R}^{100}$ ，我们在第一梯度分量中注入高斯噪声 $\mathcal{N}(0, 100^2)$ ，以模拟极端噪声稀疏性。

E.2 CNNs实验细节

在此，我们展示了图2b中省略的实验细节。我们遵循了 Bernstein 等人的代码库，该代码库可在<https://github.com/jxbz/signSGD>找到。为确保与我们的 LLM 框架进行完全标准且精确的数学比较，我们将他们的精确架构定义移植到现代 PyTorch，并使用 PyTorch 的分布式数据并行（DDP）实现了一个完全分布式的方差跟踪系统。然后，我们在四块 NVIDIA Pro 6000 GPU（每块 96GB VRAM）上运行所有实验。